



D3.2 - DESCRIPTION OF THE VALIDATION METHODOLOGY

Work package	WP3 Definition and analysis of relevant case studies for the circular bioeconomy
Task	T3.2 Consistency check between case studies and monitoring and assessment framework
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Abstract

The transition from linear fossil-based systems to circular and bio-based systems represents an opportunity and a suitable pathway for achieving several SDGs. Indeed, circular bio-based systems depict a great opportunity to reconcile sustainable long-term growth with environmental protection through the prudent use of renewable resources for industrial purposes. This needed transition is a complex process, which does not simply require innovative technologies from the supply-side, but also societal transformations based on a multi-actor process. The circular bioeconomy meta-sector may be a good candidate to put forward a new economic model, which requires transformative policies, purposeful innovation, access to finance, risk-taking capacity as well as new and sustainable business models and markets. However, a critical assessment of the environmental, social and economic impacts of the current linear fossil-based economy, as well as of the improvement potential associated with circular bio-based systems, is needed to underpin the identification of policy priorities. Bearing this in mind, SUSTRACK is a three-year project aimed at supporting policymakers in their efforts to develop sustainable pathways to replace fossil and carbon-intensive systems with sustainable circular and biobased systems (at the EU and regional scale), contributing to achieving the European Green Deal's objectives. This will be done by: identifying environmental, economic and social limits of a linear carbon-intensive and fossil-based economy; improving existing assessment methodologies; assessing the environmental, social and economic impacts of the EU's current linear fossil-based economy; comparing multiple transition scenarios focusing on the most carbon intensive sectors; identifying priorities according to scenarios analysed in the project and develop guidelines and policy recommendations.

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EXECUTIVE SUMMARY

The goal of achieving net-zero emissions by the European economy by 2050 demands a shift from linear, fossil-based systems to circular, bio-based systems. This transition is vital for meeting the United Nations Sustainable Development Goals and reconciling environmental protection with sustainable growth. However, the complexity of the transition relies on societal transformations, cutting-edge technologies, and multi-actor processes, requiring a new economic framework and policy priorities that align with the European Green Deal.

This report presents the SUSTRACK T3.2 results on the validation of the monitoring and assessment framework and guidelines for the implementation of the case studies work. In T3.1, 10 case studies were selected. In T2.1 a gap analysis was done of existing monitoring indicators and methodologies for assessing environmental, social and economic impacts of the transition from the current linear fossil-based economy to a more circular bio-based economy. In this deliverable, firstly the outcomes of a consistency check of the resulting indicator framework from T2.1 is presented per case study. This provides a further improved understanding of the indicators and metrics that are missing and/or need further elaboration in T2.2 and how the monitoring framework can be further improved as part of T2.3. This is also in the context of different case studies analysed in T3.3. This is then followed by general feedback for the further development of the monitoring and assessment framework in T2.3 in terms of clarity, completeness and other relevant aspects. Finally, guidelines are presented for mapping stakeholders, policy instruments, information and data collection for the case study analysis to be performed in T3.3. Additionally, the linkages with other SUSTRACK WP activities are defined to which the case studies analysis will provide input for the transition to a more circular biobased economy.

TABLE OF CONTENTS

1 Introduction	11
2 Approach and Methodology	12
3 Results	14
3.1 Proposed additional indicators to be incorporated in the bioeconomy monitoring and assessment framework	14
3.1.1 Proposed additional societal objectives	15
3.1.2 Proposed additional s-LCA and LCC indicators	19
3.1.3 Proposed additional indicators for regional level assessment	21
3.2 Consistency check by each case study	24
3.2.1 Construction - Cross laminated timber	24
3.2.2 Construction - Biochar as an additive in concrete	32
3.2.3 Construction - Hemp insulation	36
3.2.4 Textile - Recycled carded wool from Prato, Italy	44
3.2.5 Textile - Man-made cellulosic fibres, like Lyocell	50
3.2.6 Textile - Latxa sheep wool from the Basque Country of Spain	59
3.2.7 Chemical - Chemicals from waste	64
3.2.8 Chemical - Bio-MEG, bio-MPG, and renewable functional fillers from wood	69
3.2.9 Plastics - Polylactic acid	73
3.2.10 Plastics - Polypropylene from used cooking oil	78
3.3 General feedback on the bioeconomy monitoring and assessment framework	82
3.4 Insights generated from the validation of the bioeconomy monitoring and assessment framework	87
4. Guidance for Case Study Analysis	89
4.1 Mapping stakeholders	90
4.2 Mapping policies of relevance	91
4.3 Information and data collection from literature and stakeholders	92

4.4 Information needs for other Work packages for case studies	92
4.4.1 Linkage with WP2 related to gap analysis and closing the gaps	92
4.4.2 Linkage with WP4 on pathways for changing and defining potential (policy) intervention options	94
4.4.3 Linkage with WP5 on relevant policy instruments	94
4 Conclusions and further steps in WP3	95
References	97

LIST OF FIGURES

Figure 1: Overview of the stepwise approach applied for each case study for consistency check of the bioeconomy monitoring and assessment framework	12
Figure 2: Bioeconomy system made up of different stakeholders	91
Figure 3: Overview of the linkage of WP2 and WP3 in the SUSTRACK project	93

LIST OF TABLES

Table 1: Selected case studies and corresponding sectors	11
Table 2: Proposed 3 additional societal objectives with its linked sub-topics and indicators	15
Table 3: Proposed addition of s-LCA and LCC indicators at product level	19
Table 4: Proposed indicators for regional level assessment	22
Table 5: Consistency check for the Construction - Cross laminated timber case study	25
Table 6: Highly relevant indicators identified for the Construction - Cross laminated timber case study	26
Table 7: Consistency check for the Construction - Biochar as an additive in concrete case study	32
Table 8: Highly relevant indicators identified for the Construction - Biochar as an additive in concrete case study	34
Table 9: Consistency check for the Construction - Hemp insulation case study	37
Table 10: Highly relevant indicators identified for the Construction - Hemp insulation case study	38
Table 11: Consistency check for the Textile - Recycled carded wool from Prato, Italy case study	44
Table 12: Highly relevant indicators identified for the Textile - Recycled carded wool from Prato, Italy case study	46
Table 13: Consistency check for the Textile - Man-made cellulosic fibres, like Lyocell case study	50
Table 14: Highly relevant indicators identified for the Textile - Man-made cellulosic fibres, like Lyocell case study	52

Table 15: Consistency check for the Textile - Latxa sheep wool from the Basque Country of Spain case study	59
Table 16: Highly relevant indicators identified for the Textile - Latxa sheep wool from the Basque Country of Spain case study	61
Table 17: Consistency check for the Chemical - Chemicals from waste case study	64
Table 18: Highly relevant indicators identified for the Chemical - Chemicals from waste case study	66
Table 19: Consistency check for the Chemical - Bio-MEG, bio-MPG, and renewable functional fillers from wood case study	69
Table 20: Highly relevant indicators identified for the Chemical - Bio-MEG, bio-MPG, and renewable functional fillers from wood case study	71
Table 21: Consistency check for the Plastics - Polylactic acid case study	74
Table 22: Highly relevant indicators identified for the Plastics - Polylactic acid case study	75
Table 23: Consistency check for the Plastics - Polypropylene from used cooking oil case study	78
Table 24: Highly relevant indicators identified for the Plastics - Polypropylene from used cooking oil case study	80
Table 25: Consolidated and categorised general feedback on the bioeconomy monitoring and assessment framework	83
Table 26: Research gaps identified in T2.1 to be considered to be addressed in case studies	93
Table 27: Information to be collected regarding relevant policy instruments	95

List of Abbreviations	
Abbreviation/Acronym	Description
CBBE	Circular, bio-based economy
CLT	Cross-Laminated Timber
EC	Environmental Commission
EU	European Union
GEM	General Equilibrium Modelling
GHG	Greenhouse Gas
GWP	Global Warming Potential
HVO	hydrotreated vegetable oil
ISTAT	National Statistics Institute of Italy
LCA	Life Cycle Assessment
LCC	Life Cycle Costing
MEG	Monoethylene Glycol
MMCFs	Man-made cellulosic fibres
MPG	Monopropylene glycol
PLA	Polylactic acid
PP	Polypropylene
R&D	Research and Development
s-LCA	Social Life Cycle Assessment
SDG	Sustainable Development Goal
SO	Societal Objective
T	Task
UCO	Used cooking oil
WP	Work package

1 INTRODUCTION

The SUSTRACK project adopts a bottom-up approach to the development and validation of key project outputs (including the Bioeconomy Assessment and Monitoring Framework developed in WP2) by testing them against real case studies. In T3.1, 10 case studies were selected from four sectors that face critical challenges (see Table 1) within the current linear system: (a) the construction sector (requiring, e.g., a shift from concrete to sustainable renewable alternatives); (b) the textile sector (requiring, e.g., a shift from fossil-based to bio-based fibres); (c) the (fine) chemical sector (requiring, e.g., a transition to renewable carbon sources and biorefineries); and (d) the plastics sector (requiring, e.g., a shift from fossil-based to bio-based polymers).

Table 1: Selected case studies and corresponding sectors

Sector	Case study
Construction	Cross-laminated timber
Construction	Biochar as an additive in concrete
Construction	Hemp insulation
Textile	Recycled carded wool from Prato, Italy
Textile	Man-made cellulosic fibres, like Lyocell
Textile	Latxa sheep wool from the Basque Country of Spain
Chemical	Chemicals from waste
Chemical	Bio-MEG, bio-MPG, and renewable functional fillers from wood
Plastics	Polylactic acid (PLA)
Plastics	Polypropylene from used cooking oil

The objective of T3.2 is to validate the applicability and comprehensiveness of the bioeconomy monitoring and assessment framework developed in WP2. This framework includes a database of indicators structured under societal objectives and its sub-topics in line with the approach used in Kardung et al. (2021). For this, the 5 societal objectives of the EU Bioeconomy strategy (EC, 2018) were taken as basis i.e., (1) Food and nutrition security; (2) Sustainable natural resource management; (3) Dependence on non-renewable resources; (4) Mitigating and adapting to climate change; (5) Employment and economic competitiveness.

In T3.2, the specific 10 case studies and their needs are matched to relevant indicators following a stepwise approach and, any gaps - in terms of information available on existing indicators or additional indicator requirements - are identified. This will provide feedback for WP2 to adapt and generate additional guidance on existing indicators in the monitoring framework as well as complement the framework with additional indicators that were found relevant. Additionally, this task conducts a review of the framework in general and provides feedback for further development in T2.3. This includes consideration of for example: the categorization of sub-topics within societal objectives, possible overlap between indicators, and clarity of information provided in the framework.

The insights generated from the consistency checks carried out for each case study are reported and can provide information on more prominent indicators from the framework and identified

indicator gaps. This will also provide a linkage of the case study assessment to T2.2 for addressing the gaps identified and for further improvement of the framework in T2.3.

T3.2 also provides guidance for case study analysis to be carried out in T3.3. This considers the mapping of stakeholders and policies of relevance for case studies. Moreover, it provides guidance in terms of information and data collection. This refers to identifying and reviewing relevant (publicly) available information (e.g., publications, reports) and the results of previous environmental, social, and economic assessments. This also considers engagement with stakeholders through emails, interviews or workshops organised to generate additional knowledge on the case studies.

The report is structured as follows: Chapter 2 presents the methodological approach used to validate the bioeconomy monitoring and assessment framework; Chapter 3 presents the results from the consistency check carried out for each of the case studies and derived general insights; Chapter 4 provides the guidance for the case study assessment for stakeholder and relevant policy mapping, information collection from literature and stakeholder, and linkage with other work packages. Finally, Chapter 5 reports the main conclusions from this task and provides information on the next steps in WP3.

2 APPROACH AND METHODOLOGY

In T2.3, a preliminary monitoring and assessment framework was developed, containing a list of indicators structured under Bioeconomy strategy societal objectives (SOs) and its sub-topics as described above, to be utilized for mapping the effects of sustainability interventions in the transition to circular, bio-based systems. This was provided as an Excel file as input for this task. In T3.2, the degree of applicability and comprehensiveness of this preliminary proposed framework was checked for each of the 10 case studies identified in T3.1, belonging to the four critical sectors of interest to the SUSTRACK project. Simultaneously, this task concerned a first effort in the iterative process aimed at gaining a better understanding of the relevant indicators for the case studies. The stepwise approach applied is illustrated in Figure 1.

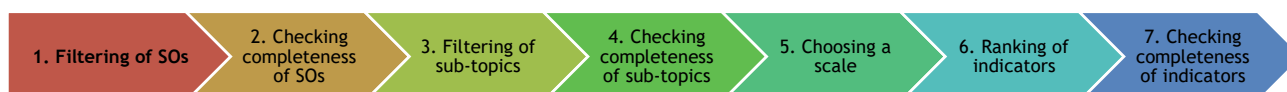


Figure 1: Overview of the stepwise approach applied for each case study for consistency check of the bioeconomy monitoring and assessment framework

To begin with, a copy of the preliminary monitoring and assessment framework Excel file (please see the supplementary material for D2.1) was made for each of the case studies and stored under the case study title. The following steps were followed in these Excel files created for each case study:

1. Filtering of societal objectives (SOs)

For the context of this case study, the societal objectives that are relevant are selected and the not relevant are filtered out.

2. Checking completeness of SOs

A check is performed to see if there are any societal objectives that are also relevant but are not included in this list. If this is the case, additional SOs are proposed, in order to ensure a sufficiently complete coverage of SOs deemed relevant in light of the examined case study.

3. Filtering of sub-topics

A second filtering step is performed to determine which sub-topics, linked to the SOs selected to be relevant in step 1, are considered applicable to the examined case study.

4. Checking completeness of sub-topics

Similar to step 2, a check is performed to see if there are any sub-topics linked to any of the relevant SOs which would be relevant for the analysis of this case study are missing from the framework. If this is the case, suggestions for additional sub-topics are made.

5. Choosing a scale

For the context of this case study, filtering is made based on which scale of assessment applies to that case study i.e., product-level, macro-level (i.e., country or region), or a mix of both.

6. Ranking of indicators

The completion of the above 5 steps results in a list of indicators remaining following the filtering performed. In this step, the evaluation of the indicators is done in terms of the relevance and suitability for this case study. The indicators are marked as either of “high”, “medium”, “low”, or “no” relevance. If sufficient information was not available to judge the relevance of the indicator at this stage, a “?” was also possible to assign to be further elaborated in the next steps.

7. Checking completeness of indicators

The indicators considered highly relevant are then added in a separate table for reporting purposes. In the last step, this list of highly relevant indicators is reviewed to see if any additional indicators would be needed to assess this case study. If this is the case, additional indicators are proposed and the list is extended with these additionally identified indicators. These additional indicators are marked red for differentiation.

A template was prepared based on this stepwise approach and shared with the case study owners to facilitate the reporting of their analysis. This reporting template is aimed at providing an overview of the activities performed for each case study in their Excel files and the resulting indicators identified to be of high relevance. Based on their current level of knowledge of the case study, each case study owner performed the consistency check exercise and filled the template for their case study. This reporting done per case study is provided in Section 3.2. It

should be noted that the case study owners had at this point different levels of knowledge on the case studies that they were responsible for. For example, for the Textile-Latxa case study, the case study owner Tecnalia had extensive information on the case study and its sustainability performance from the recent Orienting project¹ where they performed a Life Cycle Assessment (LCA), Social Life Cycle Assessment (s-LCA) and Life Cycle Costing (LCC) for this case study. For some other case studies, the case study owners already had an initial literature review on the available information on the value chain, relevant stakeholders and sustainability performance for the case study at hand. This was the case for example for the Chemicals from waste study, Construction - biochar study and Textile - Prato study. Whereas, for the other case studies, expert judgement was used with the level of information the case study owner had at this point for carrying out the consistency check and evaluation of the relevance of the indicators.

The identification of the indicators that are relevant to the case studies is considered to be an iterative process. As part of T3.2, a first review is conducted and presented in Section 3.1. This list of indicators will be further elaborated and refined as part of the T3.3 case study assessment where the case study owners will collect data and information on their case studies through literature review and stakeholder engagement. In this next stage, the data availability to assess these indicators will be analysed. Furthermore, in filling the gaps identified, close collaboration will be made with T2.2 development in terms of further development of indicators/methods and their testing in relevant case studies.

In addition to the consistency check exercise and the initial identification of the highly relevant indicators for each case study, the execution of the consistency check exercise was utilized to collect generic feedback on the preliminary monitoring and assessment framework that will be used as input in T2.3 for further improvement and optimization of the framework. These general remarks were collected and reported separately in Section 3.3.

3 RESULTS

Firstly, the additional indicators proposed to be incorporated in the bioeconomy monitoring and assessment framework are provided in Section 3.1. The results of the consistency check carried out for each case study is presented in Section 3.2 (3.2.1 through 3.2.10) which provide the analysis of the most relevant indicators identified at this stage of the case study analysis as well as the gaps identified and suggestion of additions/edits to further improve the bioeconomy monitoring and assessment framework to better assess the case study. Furthermore, general feedback on the bioeconomy monitoring and assessment framework is collected from the case studies and presented in Section 3.3. The insights generated from the analysis of the results reported are presented in Section 3.4.

3.1 Proposed additional indicators to be incorporated in the bioeconomy monitoring and assessment framework

¹ <https://orienting.eu/>

The consistency check carried out at a higher level resulted in the proposal of 3 additional sets of indicators to be incorporated in the bioeconomy monitoring and assessment framework. The motivation to include each of these indicator sets and the sources used for identifying them is described in the following subsections. The first set concerns the proposal for 3 additional societal objectives with their linked sub-topics and indicators. With the second set, it is proposed to include in the monitoring framework s-LCA and LCC indicators relevant to assess the social and economic performance at a product level. Thirdly, to better assess case studies that have a regional dimension, additional indicators are proposed to be included. These proposed indicators were reviewed in individual case study consistency checks if additional indicators were to be proposed (related to step 7 of the stepwise approach).

3.1.1 Proposed additional societal objectives

Eventually, 3 additional societal objectives with their linked sub-topics and indicators have been considered based on the literature review of indicators carried out in T2.1. These are:

- a. Disease prevention and human health
- b. Just inclusive and resilient economy
- c. Risk assessment and monitoring

These societal objectives were deemed relevant within T2.1 after (1) reviewing the work of Bracco et al. (2019), who proposed a scheme of indicators at the territorial and product levels to monitor and evaluate the sustainability of bioeconomy, and (2) cross-checking their coverage/representativeness in the current SOs promoted by the EU Bioeconomy Strategy. On the one hand, as the EU bioeconomy monitoring framework is mainly thought for territories (countries), it does not address, to a great extent, purely health issues or risk assessment measures, which in turn are crucial when concentrating at the product level. On the other hand, the EU bioeconomy strategy is strongly oriented towards strengthening EU competitiveness and job creation but fails to include key aspects such as inclusion and resilience. Table 2 presents the 3 additional SOs along with their linked sub-topics and indicators.

Table 2: Proposed additional societal objectives with their linked sub-topics and indicators

#	Societal objectives	Sub-topics	Indicators	Domains	Scale
6	Disease prevention and human health	Food safety	Quality or number of information/signs on product health and safety	Society	Product
6	Disease prevention and human health	Food safety	Number of consumer complaints	Society	Product
6	Disease prevention and human health	Food safety	Quality of labels of health and safety requirements	Society	Product
6	Disease prevention and human health	Food safety	Total number of incidents of non-compliance with regulations and	Society	Product

			voluntary codes concerning health and safety impacts of products and services and type of outcomes		
6	Disease prevention and human health	Disease/hazards prevention (in biomass production and processing)	Rate of occupational injury, illness and fatalities / final product unit	Society	Product
6	Disease prevention and human health	Disease/hazards prevention (in biomass production and processing)	Hours of training on health and safety hazards as defined in national law and international standards	Society	Product
6	Disease prevention and human health	Disease/hazards prevention (in biomass production and processing)	% of crop supply came from health and safety low-risk countries with corrective actions taken for any known high-risk sites (level of risk of the country is calculated according to Amfori Country Risk Classification)	Society	Product
6	Disease prevention and human health	Disease/hazards prevention (in biomass production and processing)	% of crop supply came from health and safety high-risk countries that have high-risk sites for which we took corrective actions (level of risk of the country is calculated according to Amfori Country Risk Classification)	Society	Product
6	Disease prevention and human health	Human health	Human toxicity and cancer effects (Comparative Toxic Unit for humans, CTUh)	Society	Product
6	Disease prevention and human health	Human health	Human toxicity - non-cancer effects (Comparative Toxic Unit for humans, CTUh)	Society	Product
6	Disease prevention and human health	Human health	Ionising radiation / product unit (kg eq U235)	Society	Product
6	Disease prevention and human health	Human health	Particulate matter as respiratory inorganics / product unit (kg eq PM2.5)	Society	Product
6	Disease prevention and human health	Human health	% of hazardous (carcinogenic, reprotoxic, mutagenic) substances for human health/product unit	Society	Product
6	Disease prevention and human health	Human health	Amount of genetically modified micro-organisms or any micro-organisms that pose a risk to human health is released outside	Society	Product

			the processing (µg/production facility)		
6	Disease prevention and human health	Human health	Ozone Depletion Potential (ODP) / product unit (kg eq. Chlorofluorocarbons (CFC)-11)	Society	Product
7	Just inclusive and resilient economy	Gender equality	Number of women having equal access to training and education and equal access to products and services as men	Society	Product
7	Just inclusive and resilient economy	Gender equality	Number of women having equal job opportunities as men	Society	Product
7	Just inclusive and resilient economy	Gender equality	Ratio of basic salary of men to women by employee category	Society	Product
7	Just inclusive and resilient economy	Inclusiveness	% of smallholder farmer sourced crop supply, by mass, was sourced from smallholder farmers that are supported by a program to increase opportunities for agricultural training, inputs, and services	Society	Product
7	Just inclusive and resilient economy	Inclusiveness	Hours of training to potentially less-advantaged group members, those in remote areas, and those with limited literacy	Society	Product
7	Just inclusive and resilient economy	Inclusiveness	Support to vulnerable people (yes/no)	Society	Product
7	Just inclusive and resilient economy	Physical infrastructure	Presence of necessary infrastructure for safe burning of processing waste and by-products (yes/no)	Economy	Product
7	Just inclusive and resilient economy	Physical infrastructure	Infrastructure and logistics for distribution of bioproducts (yes/no)	Economy	Product
7	Just inclusive and resilient economy	Resilience of rural communities - social protection	Number of co-operatives and micro-credit schemes that support empowerment of small-scale farmers and rural communities	Society	Product
7	Just inclusive and resilient economy	Resilience of rural communities - social protection	Number of operations to optimise local employment	Society	Product

8	Risk assessment and monitoring	Risk assessment	Risks for negative impacts on price and supply of fuelwood (yes/no)	Society	Product
8	Risk assessment and monitoring	Risk assessment	Strength of organizational risk assessment with regard to potential for material resource conflict (yes/no)	Society	Product
8	Risk assessment and monitoring	Monitoring and accountability systems	Presence of a system that ensures that all forms of bribery, conflicts of business interest and fraudulent practices are prohibited, including a written policy by the management and appropriate staff training (yes/no)	Society	Product
8	Risk assessment and monitoring	Monitoring and accountability systems	Presence of a register containing all evidence of legal compliance (e.g. permits, licenses, evidence of lease, concessions, etc.) and a system ensuring that auxiliary conditions are met (yes/no)	Society	Product
8	Risk assessment and monitoring	Monitoring and accountability systems	Presence of conflict management mechanisms (yes/no)	Society	Product
8	Risk assessment and monitoring	Monitoring and accountability systems	Presence of an integrated farm planning and management system that effectively addresses environmental and social compliance and risk (yes/no)	Society, environmental	Product
8	Risk assessment and monitoring	Monitoring and accountability systems	Management effort to minimize use of hazardous substances (yes/no)	Environmental	Product
8	Risk assessment and monitoring	Monitoring and accountability systems	Presence of a legal register or equivalent system with all relevant applicable international, national and regional laws and regulations (yes/no)	Society	Product
8	Risk assessment and monitoring	Monitoring and accountability systems	Presence of a training system ensuring that personnel are aware of the laws and regulations and have access to the legal register (yes/no)	Society	Product
8	Risk assessment and monitoring	Monitoring and accountability systems	Presence of a mechanism/system to receive, respond to, and document feedback, complaints and grievances from customers, workers and communities (yes/no)	Society	Product

8	Risk assessment and monitoring	Monitoring and accountability systems	Presence of a tracking system of the amounts of sustainable material sourced and sold in order to prevent multiple claiming (yes/no)	Environmental	Product
8	Risk assessment and monitoring	Monitoring and accountability systems	Number of personnel and amount of budget allocated to implement and continuously monitor compliance with international standards	Environmental	Product
8	Risk assessment and monitoring	Monitoring and accountability systems	Number of fulfilled existing regulations on health and safety	Society, environmental	Product

3.1.2 Proposed additional s-LCA and LCC indicators

During the literature review carried out in T2.1, it was found out that differently from the environmental dimension, which was well represented by the indicators provided in Bracco et al. (2019), the economic and social dimension was underrepresented. Hence, building on the recent work of the ORIENTING project² the list of life cycle indicators covering the social and economic pillars was considered. These s-LCA and LCC indicators for social and economic assessment at the product level are presented in Table 3. While the economic indicators fall within the existing SO5 Employment and economic competitiveness, the social indicators are mostly linked with the newly proposed SOs of SO6 Disease prevention and human health and SO7 Just inclusive and resilient economy.

Table 3: Proposed additional s-LCA and LCC indicators at the product level (s-LCA are shaded yellow, LCC are shaded blue)

#	Societal objectives	Sub-topics	Indicators	Domains	Scale
5	Employment and economic competitiveness	Bioeconomy-driving Policies	Labour Laws/Conventions	Society	Product
5	Employment and economic competitiveness	Employment	Unemployment	Society	Product
6	Disease prevention and human health	Human health	Occupational Tox & Haz	Society	Product
6	Disease prevention and human health	Human health	Injuries & Fatalities	Society	Product
6	Disease prevention and human health	Human health	Non-Communicable Diseases	Society	Product
6	Disease prevention and human health	Human health	Communicable Diseases	Society	Product

² See ORIENTING (2022) Deliverable D2.3 'LCSA methodology to be implemented in WP4 demonstrations' for further information

7	Just inclusive and resilient economy	Labour rights	Wage	Society	Product
7	Just inclusive and resilient economy	Labour rights	Child Labour	Society	Product
7	Just inclusive and resilient economy	Labour rights	Forced Labour	Society	Product
7	Just inclusive and resilient economy	Labour rights	Excessive Work Time	Society	Product
7	Just inclusive and resilient economy	Labour rights	Freedom of Association	Society	Product
7	Just inclusive and resilient economy	Labour rights	Migrant Labour	Society	Product
7	Just inclusive and resilient economy	Labour rights	Social Benefits	Society	Product
7	Just inclusive and resilient economy	Labour rights	Discrimination	Society	Product
7	Just inclusive and resilient economy	Respect of indigenous rights and land rights	Indigenous Rights	Society	Product
7	Just inclusive and resilient economy	Gender equality	Gender Equity	Society	Product
7	Just inclusive and resilient economy	Respect of indigenous rights and land rights	High Conflict Zones	Society	Product
7	Just inclusive and resilient economy	Access to basic needs	Access to Drinking Water	Society	Product
7	Just inclusive and resilient economy	Access to basic needs	Access to Sanitation	Society	Product
7	Just inclusive and resilient economy	Access to basic needs	Children out of School	Society	Product
7	Just inclusive and resilient economy	Access to basic needs	Access to Hospital Beds	Society	Product
7	Just inclusive and resilient economy	Inclusiveness	Smallholder Commercial Farms v	Society	Product
8	Risk assessment and monitoring	Monitoring and accountability systems	Legal System	Society	Product
8	Risk assessment and monitoring	Monitoring and accountability systems	Corruption	Society	Product
5	Employment and economic competitiveness	Comparative advantage	Capital Costs	Economy	Product
5	Employment and economic competitiveness	Comparative advantage	Raw Material/Utility costs	Economy	Product

5	Employment and competitiveness	economic	Employment	Personnel	Economy	Product
5	Employment and competitiveness	economic	Comparative advantage	Transport	Economy	Product
5	Employment and competitiveness	economic	Comparative advantage	Life cycle cost (or total undiscounted cost)	Economy	Product
5	Employment and competitiveness	economic	Environmental costs	Emission, discharge & waste related costs	Economy	Product
5	Employment and competitiveness	economic	Externalities	Soon-to-be internalised external costs	Economy	Product
5	Employment and competitiveness	economic	Externalities	Non-internalised external costs	Economy	Product
5	Employment and competitiveness	economic	Productivity	Positive cash flow	Economy	Product
5	Employment and competitiveness	economic	Comparative advantage	Net present value	Economy	Product

3.1.3 Proposed additional indicators for regional level assessment

Indicators for regional level assessment are required to perform the consistency checks for some case studies, e.g., the Prato case study for the textile sector. The report published by the National Statistics Institute of Italy (ISTAT, 2023), which is the official body for producing and disseminating official statistics in Italy, was selected as the baseline for the identification of new regional indicators in addition to the preliminary list of indicators. The ISTAT analysis was relevant to this consistency check because it provides comprehensive and reliable data on various aspects of the Italian context, including economic, social and environmental dimensions.

The identified indicators were then classified under the previously determined SOs and sub-topics. Additionally, an additional SO was proposed, namely, “Promoting global governance and cooperation”. The benefit of using this SO is twofold. First, Prato is an industrial centre with significant international links, particularly in the textile and fashion industry. Global governance and collaboration are essential to manage the complex supply chains and transnational challenges faced by this industry, such as labour rights, environmental sustainability and ethical sourcing. Second, the inclusion of this SO reflects the interconnectedness of the Prato region with the global economy and society. By promoting global governance and collaboration, we aim to address the challenges and opportunities arising from Prato's global engagement and ensure sustainable and responsible practices within the region.

Regarding the SOs identified for other case studies, especially, “Inclusive and resilient economy” was considered relevant also for the Prato case study. This is in line with Prato's efforts to promote economic growth that benefits all members of the community and promotes economic equity and resilience of rural communities involved in the textile sector. The full list of proposed indicators at the regional level are provided in Table 4.

Table 4: Proposed indicators for regional level assessment

#	Societal objectives	Sub-topics	Indicators	Domains	Scale
2	Sustainable natural resource management	Waste generation and management	Percentage of biological materials (such as wool) used in the textile industry in the region compared to total materials used.	Environment	Region
2	Sustainable natural resource management	Sustainable resource use	Water consumption per kilogram of textiles produced in the textile industry in the region	Environment	Region
2	Sustainable natural resource management	Sustainable resource use	Material extracted compared to economic value added (kg/€), OR reduction of CO2 emissions per unit of textile production compared to previous years.	Environment	Region
2	Sustainable natural resource management	Sustainable resource use	Consumption of non-renewable resources (e.g. oil, natural gas) in the regional textile industry compared to total production.	Environment	Region
2	Sustainable natural resource management	Sustainable resource use	Percentage of recycled textile materials used in the regional textile production.	Environment	Region
2	Sustainable natural resource management	Sustainable resource use	"Extension of natural areas protected or restored within the regional textile industry.	Environment	Region
2	Sustainable natural resource management	Sustainable resource use	Use of textile materials derived from sustainably managed forests (e.g. wool from sheep fed sustainable feed).	Environment	Region
2	Sustainable natural resource management	Sustainable resource use	Extent of converted or degraded agricultural land due to the expansion of the regional textile industry.	Environment	Region
2	Sustainable natural resource management	Biodiversity	Number of local threatened species protected or preserved in relation to the activities of the regional textile industry.	Environment	Region
3	Dependence on non-renewable resources	Supply security	Percentage of energy from renewable sources used in the regional textile industry.	Environment	Region
5	Employment and economic competitiveness	Innovation	N° of new sustainable technologies integrated into the regional textile industry	Economy	Region

5	Employment and economic competitiveness	Innovation	Research and development investments in the textile industry in the region(in relation to total turnover)	Economy	Region
5	Employment and economic competitiveness	Innovation	Investments in sustainable infrastructure for the regional textile industry (e.g. water recycling plants, waste treatment plants)	Economy	Region
5	Employment and economic competitiveness	Innovation	number of sustainable practices adopted by the regional textile industry and their integration into corporate relationships.	Economy	Region
5	Employment and economic competitiveness	Innovation	% of public supplies by the regional textile industry that respect sustainability criteria.	Economy	Region
7	Just inclusive and resilient economy	Resilience of rural communities - social protection	Access of rural communities involved in the regional textile industry to basic services such as drinking water, education and health services.	Society	Region
7	Just inclusive and resilient economy	Resilience of rural communities - social protection	Literacy rate and access to education in rural communities involved in the textile industry in the region	Society	Region
7	Just inclusive and resilient economy	Resilience of rural communities - social protection	Extent of protected or restored water-related ecosystem areas near the regional textile industry	Environment	Region
7	Just inclusive and resilient economy	Resilience of rural communities - social protection	Average income of families in rural communities involved in the textile industry of the region	Economy	Region
7	Just inclusive and resilient economy	Resilience of rural communities - social protection	Financial support provided to small textile businesses in the region to promote innovation.	Economy	Region
7	Just inclusive and resilient economy	Monitoring and accountability systems	Quantity of textile waste generated compared to total production in the regional textile industry.	Economy	Region
7	Just inclusive and resilient economy	Resilience of rural communities -	Investments in renewable energy technologies or infrastructure in the regional textile industry.	Economy	Region

		social protection			
7	Just inclusive and resilient economy	Resilience of rural communities - social protection	Number of educational programs or initiatives on climate change mitigation implemented in the regional textile industry.	Society	Region
7	Just inclusive and resilient economy	Resilience of rural communities - social protection	Involvement of rural communities in decisions relating to the regional textile industry.	Society	Region
7	Just inclusive and resilient economy	Inclusiveness	Availability of public information on the environmental and social impact of the textile industry in the region.	Society	Region
7	Just inclusive and resilient economy	Inclusiveness	Number of training programmes or technical cooperation of the regional textile industry with developing countries to promote sustainability.	Society	Region
9	Promoting global governance and collaboration	Partnerships and cooperation	Number of partnerships between the regional textile industry and international organisations or other global industries to promote sustainability.	Society	Region
9	Promoting global governance and collaboration	Partnerships and cooperation	Number of partnerships between the textile industry in the region and local organisations to support the development of rural communities	Society	Region

3.2 Consistency check by each case study

3.2.1 Construction - Cross laminated timber

Cross-laminated timber (CLT), also known as mass timber, is a sawn wood-based product, manufactured at a large scale. The value chain of CLT consists of several processing steps, starting with the stacking of lumber boards, which are subsequently glued and pressed in place. CLT-based wood panels are utilised in a multiplicity of predominantly construction-related applications, including in floors, roofs, and walls. As such, CLT enables developers to create wood-construction multi-floor buildings. Therewith, it is an appealing alternative to conventional building materials, like steel, concrete, and masonry. The stepwise consistency check conducted for this case study is presented in Table 5. The identified highly relevant indicators are listed in Table 6.

Table 5: Consistency check for the Construction - Cross laminated timber case study

Methodological step	Additional methodological remarks & results
1. Filtering of societal objectives	None of the SOs were excluded during the first step of the filtering process. In the case of CLT, the entire lifecycle, from the cultivation of wood in forests to the production of CLT as a bio-based material, and ultimately its end-of-life stage, comprises an array of important aspects contributing to its degree of sustainability. Accordingly, all five SOs were found relevant in this case study: - SO 1. Food and nutrition security - SO 2. Sustainable natural resource management - SO 3. Dependence on non-renewable resources - SO 4. Mitigating and adapting to climate change - SO 5. Employment and economic competitiveness
2. Checking completeness societal objectives	No additional SOs were identified.
3. Filtering of sub-topics	While forests used to produce wood for construction purposes can, to some extent, also be utilised for food production, this aspect has limited relevance in the case study of CLT. Also, water-based ecosystems are to a lesser degree relevant to wood production. Therefore, a set of sub-topics was excluded from further analysis. These comprise of the following: - From SO 1. Food and nutrition security: <i>Access to food, and Utilisation.</i> - From SO 3. Dependence on non-renewable resources: <i>Water quality and Marine environments' quality.</i> The sub-topics that were found highly relevant can be found in Table 6.
4. Checking completeness sub-topics	Two additional sub-topics are suggested for the Societal objective 2 "Sustainable natural resource management", being "Use of local resources" and "Local production". These sub-topics are closely tied to the climate impact of transportation. In the case of CLT, this encompasses both the transportation of lumber to CLT production plants and the shipping of manufactured CLT panels to construction sites. Increasing the utilisation of locally sourced lumber can significantly reduce the overall global warming potential (GWP) associated with CLT manufacturing. Similarly, the adoption of locally produced CLT panels reduces the transportation distance between manufacturing facilities to construction sites. Currently, the majority of global CLT output originates from Germany, Austria, Switzerland, Italy and Chzechia, utilising readily available softwood species. However, as CLT production expands into other regions, the utilisation of new

	and locally sourced wood species may become more prevalent in CLT production.
5. Choosing a scale	The CLT study links to the whole value chain of turning sawn wood coming from forests into a versatile and structural bio-based building product for the construction industry. Besides, this case study has a clear spatial component, as forests play a key role in a variety of ecosystem services worldwide. Therefore, the indicators linked to both the product-level and to the macro-level (i.e., country or regional level) were adopted in the database.
6. Ranking of indicators	From the filtering conducted in the above steps, 247 indicators were left out of a total of 264 at the start of this exercise. This number includes product-level (153) and macro-level (53) indicators, while the scale of the remaining (14) indicators is not classified. Based on a subjective evaluation, the remaining indicators in the database were scored in terms of relevance, resulting in: - 79 indicators of high relevance, - 141 indicators of low relevance, - 27 indicators of no relevance. From the 79 indicators labelled as highly relevant, most indicators are covered by the environmental dimension (63), followed by the economic (11), and societal (2) dimension. In addition, several highly relevant indicators were identified as a combination of dimensions: environmental and circular (2), or environmental, circular and economic (1). An overview of the indicators considered highly relevant, can be found in Table 6.
7. Checking completeness indicators	No additional indicators were identified.

Table 6: Highly relevant indicators identified for the Construction - Cross laminated timber case study

	Societal objectives	Sub-topics	Indicators	Domains	Scale
2	Sustainable natural resource management	Biodiversity	% of biomass produced on land that is/was highly biodiverse grassland	Environment	Product
2	Sustainable natural resource management	Biodiversity	% of biomass produced on land that is/was a protected area	Environment	Product
2	Sustainable natural resource management	Biodiversity	Functional diversity (Number and variety of the elements of biodiversity that influence how ecosystems function)	Environment	Product

2	Sustainable natural resource management	Land use change (LUC)	% of biomass obtained from land with high carbon stock (e.g. wetlands)	Environment	Product
2	Sustainable natural resource management	Land use change (LUC)	% of biomass obtained from land that is/was peatland	Environment	Product
2	Sustainable natural resource management	Land use change (LUC)	% of crop supply, by mass, obtained from land that is/was primary forest or other wooded land	Environment	Product
2	Sustainable natural resource management	Land use change (LUC)	Change in wood/wood fibre demand for forest products	Environment	Product
2	Sustainable natural resource management	Land use change (LUC)	Forest converted to cropland	Environment	Product
2	Sustainable natural resource management	Primary Biomass production	Wood consumption	Environment	Product
2	Sustainable natural resource management	Sustainable resource use	Intensity of fossil fuel use (m3/ton of product)	Environment	Product
2	Sustainable natural resource management	Sustainable resource use	Change in consumption level of fossil fuel resources / product unit	Environment	Product
2	Sustainable natural resource management	Sustainable resource use	% of biological content/ product	Environment	Product
2	Sustainable natural resource management	Sustainable resource use	% of carbon content derived from renewable raw materials	Environment	Product
2	Sustainable natural resource management	Energy efficiency	Primary energy savings (kWh or % of total primary energy consumed)	Environment	Product
2	Sustainable natural resource management	Energy efficiency	Energy Intensity (product value added / mJ primary energy)	Environment	Product
2	Sustainable natural resource management	Energy efficiency	Energy consumption by activity/ production process/unit of production (kWh/production process or product unit)	Environment	Product
2	Sustainable natural resource management	Forest quality	% of crop supply, by mass, was grown on fields with zero conversion High Carbon Stock (HCS) forests	Environment	Product

2	Sustainable natural resource management	Forest quality	% of crop supply, by mass, was grown on fields with zero conversion of High Conservation Value (HCV) forests	Environment	Product
2	Sustainable natural resource management	Forest quality	% of the total amount of wood used comes from sustainable forest management (economically viable, environmentally sound, socially responsible) or are waste wood according to waste wood categories	Environment	Product
2	Sustainable natural resource management	Forest quality	% of wood, wood-based, cork, cork-based, bamboo, bamboo-based materials originate from genetically modified organisms (GMO)	Environment	Product
2	Sustainable natural resource management	Forest quality	Share of forests certified for sustainable management	Environment	Product
2	Sustainable natural resource management	Forest quality	% of deforestation or afforestation /product unit	Environment	Product
2	Sustainable natural resource management	Waste generation and management	Waste generated	Environment	Product
2	Sustainable natural resource management	Waste generation and management	Procedures for separating and using recyclable materials from the waste stream (yes/no)	Environment	Product
2	Sustainable natural resource management	Waste generation and management	Presence of measures that can help to reduce waste produced by the seaweed biomass production (yes/no)	Environment	Product
2	Sustainable natural resource management	Waste generation and management	Declaration of end-of-life disposal (in the CSR report) (yes/no)	Environment	Product
2	Sustainable natural resource management	Waste generation and management	% of recycled materials for packaging	Environment	Product
2	Sustainable natural resource management	Waste generation and management	Procedures for recovering materials for other uses, such as incineration for raising process steam or heating, or agricultural use (yes/no)	Environment	Product
2	Sustainable natural resource management	Waste generation and management	% of recycled fibre raw material is used	Environment	Product
2	Sustainable natural resource management	Waste generation and management	% of product is actively being recovered and recycled	Environment	Product

2	Sustainable natural resource management	Waste generation and management	% of the total amount of wood used is waste wood according to waste wood categories	Environment	Product
2	Sustainable natural resource management	Waste generation and management	Recycling rate/cascading indexes	Environment	Product
2	Sustainable natural resource management	Waste generation and management	% of biodegradability	Environment	Product
2	Sustainable natural resource management	Waste generation and management	Post-consuming recycling rate/cascading indexes	Environment	Product
2	Sustainable natural resource management	Bio-energy replacing non-renewable energy	Change of biomass demand for energy use	Environment	Product
2	Sustainable natural resource management	Sustainable resource use	Forest footprint	Environment	Country
2	Sustainable natural resource management	Indirect Land use change (iLUC)	Loss of agricultural or forest area	Environment	Country/region
2	Sustainable natural resource management	Land use change (LUC)	Loss of agricultural or forest area	Environment	Country/region
2	Sustainable natural resource management	Biodiversity	Forest biodiversity	Environment	Country
2	Sustainable natural resource management	Land cover/use	Forest area	Environment	Country
2	Sustainable natural resource management	Sustainable resource use	Sustainable forestry	Environment	Country
3	Dependence on non-renewable resources	Bio-energy replacing non-renewable energy	% of biofuels produced with co-products, residues and waste	Environment	Product
3	Dependence on non-renewable resources	Material efficiency	Carbon intensity	Environment	Product
3	Dependence on non-renewable resources	Material efficiency	Material and (municipal) waste recycling and recovery rates	Economy, environment, circularity	Country
3	Dependence on non-renewable resources	Bio-energy replacing non-renewable energy	Biofuels	Environment	Country

3	Dependence on non-renewable resources	Bio-energy replacing non-renewable energy	Post-consumer product for energy	Environment	Country
3	Dependence on non-renewable resources	Bio-energy replacing non-renewable energy	Industrial residues used in industrial processes for energy	Environment	Country
3	Dependence on non-renewable resources	Bio-material replacing non-renewable resources	Wood-based constructions	Environment	Country
3	Dependence on non-renewable resources	Certified bio-based products	Number	Environment	Country
3	Dependence on non-renewable resources	Certified bio-based products	Bio-based content	Environment	Country
3	Dependence on non-renewable resources	Certified bio-based products	Consumption (or share consumption)	Environment	Country
3	Dependence on non-renewable resources	Material efficiency use	Cascaded use of biomass	Environment	Country
3	Dependence on non-renewable resources	Material efficiency use	Biomass Utilisation Factor (BUF)	Environment, Circularity	Product
3	Dependence on non-renewable resources	Material efficiency use	Material circularity indicator (MCI)	Environment, Circularity	Country
4	Mitigating and adapting to climate change	Greenhouse gas emissions	Life cycle GHG emissions (gr eq. CO ₂ /product unit)	Environment	Product
4	Mitigating and adapting to climate change	Greenhouse gas emissions	Rate of GHG emissions reduction or savings	Environment	Product
4	Mitigating and adapting to climate change	Greenhouse gas emissions	Carbon stock (kg/unit of biomass)	Environment	Product
4	Mitigating and adapting to climate change	Greenhouse gas emissions	Potential leakages (gr eq. CO ₂ /product unit)	Environment	Product
4	Mitigating and adapting to climate change	Greenhouse gas emissions	Global Warming Potential (GWP100; gr eq CO ₂)	Environment	Product

4	Mitigating and to climate change	Air quality	Climate regulation potential (CRP) (ton C/m3)	Environment	Product
4	Mitigating and to climate change	Climate change adaptation	Diversity of tree species	Environment	Country
4	Mitigating and to climate change	Climate footprint	Climate footprint	Environment	Country
4	Mitigating and to climate change	Greenhouse gas emissions	Greenhouse gas emissions	Environment	Country
4	Mitigating and to climate change	Greenhouse gas emissions	Forest carbon emissions and removals	Environment	...
4	Mitigating and to climate change	Greenhouse gas emissions	Energy and industrial carbon emissions and removals	Environment	...
5	Employment and economic competitiveness	Productivity	Land for biomass production	Environmental	Product
2	Sustainable natural resource management	Sustainable resource use	Residual biomass potential	Economy	Product
2	Sustainable natural resource management	Primary Biomass production	Forests	Economy	Country
3	Dependence on non-renewable resources	Material efficiency use	(Total raw) material productivity	Economy	Product
3	Dependence on non-renewable resources	Biomass self-sufficiency rate	Forestry biomass	Economy	Country
3	Dependence on non-renewable resources	Biomass self-sufficiency rate	Biomass from waste	Economy	Country
3	Dependence on non-renewable resources	Material efficiency use	Recycling rate of bio-based products	Economy	Country
3	Dependence on non-renewable resources	Material efficiency use	Post-consumer product used in industrial processes as material	Economy	Country
3	Dependence on non-renewable resources	Material efficiency use	Industrial residues used in industrial processes as material	Economy	Country

5	Employment and economic competitiveness	Import and export of bioeconomy raw materials and products	Import and export of bioeconomy raw materials and products	Economy	Country
5	Employment and economic competitiveness	Production and consumption of non-food and feed bio-based products	Production and consumption of non-food and feed bio-based products	Economy	Country
5	Employment and economic competitiveness	Comparative advantage	Potential market share of the product	Economy	Product
5	Employment and economic competitiveness	Bioeconomy-driving Policies	Existence of (legal) obligation on public sustainability report (yes/no)	Society	Product
5	Employment and economic competitiveness	Innovation	Number of certifications/labels the organisation obtained for the product/site	Society	Product

3.2.2 Construction - Biochar as an additive in concrete

Biochar is a porous, carbon-rich material with a carbon content of up to 90%. Besides bio-oils and syngas, it is one of the three products that can be obtained from the pyrolysis of primary or secondary biomass. Due to its beneficial physical characteristics, biochar can be applied as a partial substitute of cement or filler in concrete. To that end, the biochar is mixed with coarse aggregates to function as a hydraulic binding agent. Since the cementitious composite industry is notorious for its high share in the worldwide greenhouse gas emissions, its replacement by biochar is considered as a major potential solution in the combat of climate change. The stepwise consistency check carried out for this case study is presented in Table 7. The identified highly relevant indicators are listed in Table 8.

Table 7: Consistency check for the Construction - Biochar as an additive in concrete case study

Methodological step	Additional methodological remarks & results
1. Filtering of societal objectives	Since none of the components of concrete nor biochar are produced from biomass that is suitable for food consumption, SO 1. Food and nutrition security was not considered. The other four SOs were found relevant for this case study: - SO 2. Sustainable natural resource management - SO 3. Dependence on non-renewable resources - SO 4. Mitigating and adapting to climate change - SO 5. Employment and economic competitiveness
2. Checking completeness	No additional SOs were identified.

societal objectives	
3. Filtering of sub-topics	Marine ecosystems are to a limited extent relevant in the context of the biochar production system. Therefore, the sub-topic <i>Marine environments' quality</i>, linked to SO 3. Dependence on non-renewable resources was excluded from further analysis. The sub-topics that were found highly relevant can be found in Table 8.
4. Checking completeness sub-topics	No additional sub-topics were identified.
5. Choosing a scale	The biochar in concrete case study considers the conventional, fossil-based value chain of concrete production against an alternative value chain involving the production of biochar from biomass. Biochar is typically made from waste wood but can be produced from various kinds of (waste-based) biomasses. Each value chain is linked to a specific product, which combines to make a third product. Therefore, it was decided to adopt the indicators linked to the product-level, and to deselect the macro-level (i.e., country or regional level) focused indicators in the database.
6. Ranking of indicators	From the filtering carried out in the earlier steps, 170 indicators were left out of a total initial list of 264. A literature review was performed in order to assess the environmental, economic and societal sustainability aspects of concrete and biochar production. Based on this search, the remaining indicators in the database were scored in terms of their relevance, resulting in: - 37 indicators of high relevance, - 63 indicators of low relevance, - 60 indicators of no relevance, - 10 indicators where the relevance is unknown. From the 37 indicators labelled as highly relevant, most indicators are covered by the environmental dimension (31), followed by the societal (5), and economic (1) dimension. In addition, several highly relevant indicators were identified as a combination of dimensions: environmental and circular (2), or environmental, circular and economic (1). An overview of the indicators considered highly relevant can be found in Table 8.
7. Checking completeness indicators	No additional indicators were identified.

Table 8: Highly relevant indicators identified for the Construction - Biochar as an additive in concrete case study (all product scale)

	Societal objectives	Sub-topics	Indicators	Domains
2	Sustainable natural resource management	Sustainable resource use	Biodiversity damage potential (BDP; m2*year*PAS (potentially affected species)	Environment
2	Sustainable natural resource management	Sustainable resource use	% of crop supply, by mass, obtained from land that is/was primary forest or other wooded land	Environment
2	Sustainable natural resource management	Sustainable resource use	% of biofuels and bioliquids made from raw material obtained from land that is/was continuously forested	Environment
2	Sustainable natural resource management	Sustainable resource use	Wood consumption	Environment
2	Sustainable natural resource management	Sustainable resource use	Amount of sold crop exceeding harvesting volume	Environment
2	Sustainable natural resource management	Sustainable resource use	Water purification potential through physicochemical filtration (WPPPCF; Physicochemical capacity of ecosystems to clean a polluted suspension) (yes/no)	Environment
2	Sustainable natural resource management	Sustainable resource use	Resource depletion (mineral, fossil) / product unit (MJ)	Environment
2	Sustainable natural resource management	Sustainable resource use	Intensity of fossil fuel use (m3/ton of product)	Environment
2	Sustainable natural resource management	Sustainable resource use	% of biological content/ product	Environment
2	Sustainable natural resource management	Sustainable resource use	% of carbon content derived from renewable raw materials	Environment
2	Sustainable natural resource management	Energy efficiency	Renewable energies (RES) share in final energy consumption / product unit	Environment
2	Sustainable natural resource management	Water quality	Ecotoxicity for aquatic freshwater (Comparative Toxic Unit for ecosystems)	Environment
2	Sustainable natural resource management	Forest quality	Share of forests certified for sustainable management	Environment

2	Sustainable natural resource management	Forest quality	% of deforestation or afforestation /product unit	Environment
2	Sustainable natural resource management	Hazardous substances management	% of hazardous (carcinogenic, mutagenic, reprotoxic) substances	Environment
2	Sustainable natural resource management	Hazardous substances management	% of polycyclic aromatic hydrocarbon (PAH)	Environment
2	Sustainable natural resource management	Waste generation and management	Waste generated	Environment
2	Sustainable natural resource management	Waste generation and management	Procedures for separating and using recyclable materials from the waste stream (yes/no)	Environment
2	Sustainable natural resource management	Waste generation and management	Declaration of end-of-life disposal (in the CSR report) (yes/no)	Environment
2	Sustainable natural resource management	Waste generation and management	% of product is actively being recovered and recycled	Environment
2	Sustainable natural resource management	Waste generation and management	% of materials derived from recycling or recovery of any other biomass waste	Environment
2	Sustainable natural resource management	Waste generation and management	% of materials derived from any other biomass by-products	Environment
2	Sustainable natural resource management	Waste generation and management	% of the total amount of wood used is waste wood according to waste wood categories	Environment
2	Sustainable natural resource management	Waste generation and management	Recycling rate/cascading indexes	Environment
2	Sustainable natural resource management	Waste generation and management	Amount of wastewater from processing operations is discharged into aquatic systems (m3)	Environment
2	Sustainable natural resource management	Waste generation and management	% of wastewater containing potential organic and mineral contaminants are treated or recycled to prevent negative impacts	Environment
2	Sustainable natural resource management	Waste generation and management	Practices for waste storage, treatment and disposal that do not pose health or safety risks to farmers, workers, other people, or natural ecosystems (yes/no)	Environment

3	Dependence on non-renewable resources	Bio-energy replacing non-renewable energy	% of biofuels produced with co-products, residues and waste	Environment
3	Dependence on non-renewable resources	Material use efficiency	Biomass Utilisation Factor (BUF)	Environment
4	Mitigating and adapting to climate change	Greenhouse gas emissions	Life cycle GHG emissions (gr eq. CO ₂ /product unit)	Environment
4	Mitigating and adapting to climate change	Greenhouse gas emissions	Global Warming Potential (GWP100; gr eq CO ₂)	Environment
5	Employment and economic competitiveness	Bioeconomy-driving Policies	Strategy for creating the infrastructure and systems necessary for recovering and recycling materials (yes/no)	Economy
5	Employment and economic competitiveness	Bioeconomy-driving Policies	Presence of a formal policy in place concerning health and safety of all workers, including contractors (yes/no)	Society
5	Employment and economic competitiveness	Stakeholder engagement and capacity building	Presence of a law or norm regarding transparency (yes/no)	Society
5	Employment and economic competitiveness	Employment	Risk of forced labour used for production of commodity (yes/no)	Society
5	Employment and economic competitiveness	Employment	Presence of working children under the legal age (yes/no)	Society
5	Employment and economic competitiveness	Innovation	Number of innovative social project that positively impacts the local community	Society

3.2.3 Construction - Hemp insulation

Hemp insulation is produced based on hemp wool. Hemp wool contains strong, woody fibres that are derived from the shives of the hemp plant. To manufacture hemp-based insulation material, hemp shives are defibrated, chipped, pulped and formed into mats or boards. These materials have excellent insulating properties in terms of fire-resistance, breathability, and robustness. As such, hemp insulation lends itself as a suitable, bio-based alternative to incumbent insulation materials, such as fibreglass, rock and slag wool. The stepwise consistency check carried out for this case study is presented in Table 9. The identified highly relevant indicators are listed in Table 10.

Table 9: Consistency check for the Construction - Hemp insulation case study

Methodological step	Additional methodological remarks & results
1. Filtering of societal objectives	All five SOs were found relevant in this case study, as it is believed that each of them is to some degree relevant considering the hemp insulation production system: - SO 1. Food and nutrition security - SO 2. Sustainable natural resource management - SO 3. Dependence on non-renewable resources - SO 4. Mitigating and adapting to climate change - SO 5. Employment and economic competitiveness
2. Checking completeness societal objectives	No additional SOs were identified.
3. Filtering of sub-topics	The production of hemp insulation material is believed to not substantially contribute to certain specific food-related impacts. Also, water- and forest-based ecosystems are considered to have limited relevance in the context of the examined case study. Therefore, a set of sub-topics was excluded from further analysis, comprising the following: - From SO 1. Food and nutrition security: <i>Access to food</i>, and <i>Utilisation</i>. - From SO 3. Dependence on non-renewable resources: <i>Water quality</i>, <i>Forest quality</i>, and <i>Marine environments' quality</i>. On the contrary, sub-topics considered highly relevant can be found in Table 10.
4. Checking completeness sub-topics	No additional sub-topics were identified.
5. Choosing a scale	The hemp insulation case study encompasses the entire value chain of industrial hemp, starting with its cultivation as an agricultural crop, proceeding to the processing and separation of its various components (i.e., seeds, fibres, and shives), and culminating in the production of hemp insulation, using the fibres. Besides, this case study has a clear spatial component, as biomass cultivation (in this case hemp) plays an important role in a variety of ecosystem services worldwide. Therefore, the indicators linked to both the product-level and to the macro-level (i.e., country or regional level) were adopted in the database.
6. Ranking of indicators	From the filtering carried out in the above steps, 243 indicators were left out of a total initial list of 264. This number includes product-level (146) and macro-level (51) indicators, while the scale of the remaining (16) indicators is not classified.

	Based on a subjective evaluation, the remaining indicators in the database were scored in terms of relevance, resulting in: - 84 indicators of high relevance, - 132 indicators of low relevance, - 27 indicators of no relevance. From the 84 indicators labelled as highly relevant, most are covered by the environmental dimension (67), followed by the economic (11), and societal (2) dimension. In addition, several highly relevant indicators were identified as a combination of dimensions: environmental and circular (2), environmental and economic (1), or environmental, circular and economic (1). An overview of the indicators considered highly relevant, can be found in Table 10.
7. Checking completeness indicators	No additional indicators were identified.

Table 10: Highly relevant indicators identified for the Construction - Hemp insulation case study

	Societal objectives	Sub-topics	Indicators	Domains	Scale
2	Sustainable natural resource management	Land cover/use	Agricultural area	Environment	Country
2	Sustainable natural resource management	Indirect Land use change (iLUC)	Loss of agricultural or forest area	Environment	Country/region
2	Sustainable natural resource management	Land use change (LUC)	Loss of agricultural or forest area	Environment	Country/region
2	Sustainable natural resource management	Sustainable resource use	Agricultural Land footprint	Environment	Country/region
2	Sustainable natural resource management	Biodiversity	Proportion of degraded agricultural area	Environment	Country/region
2	Sustainable natural resource management	Biodiversity	% of biomass produced on land that is/was highly biodiverse grassland	Environment	Product
2	Sustainable natural resource management	Biodiversity	% of biomass produced on land that is/was a protected area	Environment	Product
2	Sustainable natural resource management	Biodiversity	Functional diversity (Number and variety of the elements of biodiversity)	Environment	Product

			that influence how ecosystems function)		
2	Sustainable natural resource management	Land use change (LUC)	% of biomass obtained from land with high carbon stock (e.g. wetlands)	Environment	Product
2	Sustainable natural resource management	Land use change (LUC)	% of biomass obtained from land that is/was peatland	Environment	Product
2	Sustainable natural resource management	Land use change (LUC)	% of crop supply, by mass, obtained from land that is/was primary forest or other wooded land	Environment	Product
2	Sustainable natural resource management	Land use change (LUC)	Agricultural land converted to energy (or bio-based product) crop	Environment	Product
2	Sustainable natural resource management	Land use change (LUC)	% of biofuels and bioliquids made from raw material obtained from land that is/was continuously forested	Environment	Product
2	Sustainable natural resource management	Land use change (LUC)	Grassland converted to cropland	Environment	Product
2	Sustainable natural resource management	Land use change (LUC)	Forest converted to cropland	Environment	Product
2	Sustainable natural resource management	Resilience of ecosystems	Erosion regulation potential (ERP; Capacity of ecosystems to stabilise soil and to prevent sediment accumulation downstream) (mass of soil lost per unit area and time)	Environment	Product
2	Sustainable natural resource management	Sustainable resource use	Amount of water used in the value-chain	Environment	Product
2	Sustainable natural resource management	Sustainable resource use	Intensity of fossil fuel use (m3/ton of product)	Environment	Product
2	Sustainable natural resource management	Sustainable resource use	Change in consumption level of fossil fuel resources / product unit	Environment	Product
2	Sustainable natural resource management	Sustainable resource use	% of biological content/ product	Environment	Product
2	Sustainable natural resource management	Sustainable resource use	% of carbon content derived from renewable raw materials	Environment	Product

2	Sustainable natural resource management	Energy efficiency	Primary energy savings (kWh or % of total primary energy consumed)	Environment	Product
2	Sustainable natural resource management	Energy efficiency	Energy Intensity (product value added / mJ primary energy)	Environment	Product
2	Sustainable natural resource management	Energy efficiency	Energy consumption by activity/ production process/unit of production (kWh/production process or product unit)	Environment	Product
2	Sustainable natural resource management	Soil quality	Soil erosion / product unit	Environment	Product
2	Sustainable natural resource management	Soil quality	Soil nutrient balance	Environment	Product
2	Sustainable natural resource management	Hazardous substances management	% of hazardous (carcinogenic, mutagenic, reprotoxic) substances	Environment	Product
2	Sustainable natural resource management	Hazardous substances management	% of plastic (synthetic plastic material, plasticizers, PVC) contained in the product	Environment	Product
2	Sustainable natural resource management	Hazardous substances management	% of biocides	Environment	Product
2	Sustainable natural resource management	Hazardous substances management	% of glyoxal or formaldehyde chemicals	Environment	Product
2	Sustainable natural resource management	Hazardous substances management	% of flame retardants	Environment	Product
2	Sustainable natural resource management	Waste generation and management	Waste generated	Environment	Product
2	Sustainable natural resource management	Waste generation and management	Procedures for separating and using recyclable materials from the waste stream (yes/no)	Environment	Product
2	Sustainable natural resource management	Waste generation and management	Presence of measures that can help to reduce waste produced by the seaweed biomass production (yes/no)	Environment	Product
2	Sustainable natural resource management	Waste generation and management	Declaration of end-of-life disposal (in the CSR report) (yes/no)	Environment	Product

2	Sustainable natural resource management	Waste generation and management	% of recycled materials for packaging	Environment	Product
2	Sustainable natural resource management	Waste generation and management	Procedures for recovering materials for other uses, such as incineration for raising process steam or heating, or agricultural use (yes/no)	Environment	Product
2	Sustainable natural resource management	Waste generation and management	% of recycled fibre raw material is used	Environment	Product
2	Sustainable natural resource management	Waste generation and management	% of product is actively being recovered and recycled	Environment	Product
2	Sustainable natural resource management	Waste generation and management	% of the total amount of agricultural biomass used come from agricultural residues	Environment	Product
2	Sustainable natural resource management	Waste generation and management	Recycling rate/cascading indexes	Environment	Product
2	Sustainable natural resource management	Waste generation and management	% of biodegradability	Environment	Product
2	Sustainable natural resource management	Waste generation and management	Post-consuming recycling rate/cascading indexes	Environment	Product
2	Sustainable natural resource management	Bio-energy replacing non-renewable energy	Change of biomass demand for energy use	Environment	Product
2	Sustainable natural resource management	Biodiversity	Agrobiodiversity	Environment	
2	Sustainable natural resource management	Sustainable resource use	Sustainable agriculture	Environment	
2	Sustainable natural resource management	Sustainability threshold levels for Bioeconomy Technologies	Amount of sold crop exceeding harvesting volume	Environment, economy	Product
3	Dependence on non-renewable resources	Material efficiency use	Material and (municipal) waste recycling and recovery rates	Economy, environment, circularity	Country
3	Dependence on non-renewable resources	Bio-energy replacing non-renewable energy	Biofuels	Environment	Country

3	Dependence on non-renewable resources	Bio-energy replacing non-renewable energy	Post-consumer product for energy	Environment	Country
3	Dependence on non-renewable resources	Bio-energy replacing non-renewable energy	Industrial residues used in industrial processes for energy	Environment	Country
3	Dependence on non-renewable resources	Certified bio-based products	Number	Environment	Country
3	Dependence on non-renewable resources	Certified bio-based products	Bio-based content	Environment	Country
3	Dependence on non-renewable resources	Certified bio-based products	Consumption (or share consumption)	Environment	Country
3	Dependence on non-renewable resources	Material efficiency use	Cascaded use of biomass	Environment	Country
3	Dependence on non-renewable resources	Bio-energy replacing non-renewable energy	% of biofuels produced with co-products, residues and waste	Environment	Product
3	Dependence on non-renewable resources	Material efficiency use	Carbon intensity	Environment	Product
3	Dependence on non-renewable resources	Material efficiency use	Material circularity indicator (MCI)	Environment, Circularity	Country
3	Dependence on non-renewable resources	Material efficiency use	Biomass Utilisation Factor (BUF)	Environment, Circularity	Product
4	Mitigating and adapting to climate change	Greenhouse emissions gas	Life cycle GHG emissions (gr eq. CO ₂ /product unit)	Environment	Product
4	Mitigating and adapting to climate change	Greenhouse emissions gas	Rate of GHG emissions reduction or savings	Environment	Product
4	Mitigating and adapting to climate change	Greenhouse emissions gas	Carbon stock (kg/unit of biomass)	Environment	Product
4	Mitigating and adapting to climate change	Greenhouse emissions gas	Potential leakages (gr eq. CO ₂ /product unit)	Environment	Product
4	Mitigating and adapting to climate change	Greenhouse emissions gas	Global Warming Potential (GWP100; gr eq CO ₂)	Environment	Product

4	Mitigating and adapting to climate change	Air quality	Climate regulation potential (CRP) (ton C/m3)	Environment	Product
4	Mitigating and adapting to climate change	Climate change adaptation	Diversity of tree species	Environment	Country
4	Mitigating and adapting to climate change	Climate footprint	Climate footprint	Environment	Country
4	Mitigating and adapting to climate change	Greenhouse gas emissions	Greenhouse gas emissions	Environment	Country
4	Mitigating and adapting to climate change	Greenhouse gas emissions	Forest carbon emissions and removals	Environment	
4	Mitigating and adapting to climate change	Greenhouse gas emissions	Energy and industrial carbon emissions and removals	Environment	
5	Employment and economic competitiveness	Productivity	Land for biomass production	Environmental	Product
2	Sustainable natural resource management	Sustainable resource use	Residual biomass potential	Economy	Region
2	Sustainable natural resource management	Primary Biomass production	Agriculture	Economy	Country
3	Dependence on non-renewable resources	Material efficiency use	(Total raw) material productivity	Economic	Country
3	Dependence on non-renewable resources	Biomass self-sufficiency rate	Agricultural biomass	Economy	Country
3	Dependence on non-renewable resources	Biomass self-sufficiency rate	Biomass from waste	Economy	Country
3	Dependence on non-renewable resources	Material efficiency use	Recycling rate of bio-based products	Economy	Country
3	Dependence on non-renewable resources	Material efficiency use	Post-consumer product used in industrial processes as material	Economy	Country
3	Dependence on non-renewable resources	Material efficiency use	Industrial residues used in industrial processes as material	Economy	Country

5	Employment and economic competitiveness	Import and export of bioeconomy raw materials and products	Import and export of bioeconomy raw materials and products	Economy	Country
5	Employment and economic competitiveness	Production and consumption of non-food and feed bio-based products	Production and consumption of non-food and feed bio-based products	Economy	Country
5	Employment and economic competitiveness	Comparative advantage	Potential market share of the product	Economy	Product
5	Employment and economic competitiveness	Bioeconomy-driving Policies	Existence of (legal) obligation on public sustainability report (yes/no)	Society	Product
5	Employment and economic competitiveness	Innovation	Number of certifications/labels the organisation obtained for the product/site	Society	Product

3.2.4 Textile - Recycled carded wool from Prato, Italy

The city of Prato, located in the Tuscany region, has a longstanding history as a textile district with the production of carded wool from recycled materials being its core business. The production of recycled carded wool is a decade-old technology of several steps from sorting, cleaning to spinning and finally warping to yarn, which can then be further woven to create fabric. The production is chemical- and dye-free, making it an interesting substitute of or complement to conventional, synthetic fibre production. The stepwise consistency check carried out for this case study is presented in Table 11. The identified highly relevant indicators are listed in Table 12.

Table 11: Consistency check for the Textile - Recycled carded wool from Prato, Italy case study

Methodological step	Additional methodological remarks & results
1. Filtering of societal objectives	SO 1. Food and nutrition security was considered irrelevant, since the present case study mainly considers discarded wool, which does not compete with food-related applications. Accordingly, four out of five SOs were found relevant in this case study: - SO 2. Sustainable natural resource management - SO 3. Dependence on non-renewable resources - SO 4. Mitigating and adapting to climate change - SO 5. Employment and economic competitiveness
2. Checking completeness	The 3 additional SOs proposed in section 3.1.1 were found relevant to be included in the monitoring framework. Additionally, ISTAT analysis was used as input for carrying out the consistency check. It provides comprehensive and

societal objectives	reliable data on various aspects of Italy, including economic, social and environmental dimensions. It is also the official body for producing and disseminating official statistics in Italy. This analysis provides us with information on the current situation in the Prato region, which helps us to adapt our social objectives (SO) to the local context. ISTAT data allows us to identify areas where there may be gaps or areas that require additional attention. Reviewing the list of SOs used in this ISTAT analysis, one additional SO is proposed that could support the assessment at the regional level. This is SO9: "Promoting global governance and collaboration." Global governance and collaboration are essential to manage the complex supply chains and transnational challenges faced by this industry, such as labour rights, environmental sustainability, and ethical sourcing. The inclusion of this SO reflects the interconnectedness of the Prato region with the global economy and society. By promoting global governance and collaboration, it is aimed to address the challenges and opportunities arising from Prato's global engagement and ensure sustainable and responsible practices within the region.
3. Filtering of sub-topics	A set of sub-topics was excluded from further analysis since the case study does not consider the impacts related to the land use, energy-related issues, or raw material but, instead, how to valorise carded wool. Excluded sub-topics comprise: - From SO 2. Sustainable natural resource management: <i>Indirect Land use change (iLUC), Land cover/use, Land use change (LUC), Primary biomass production, Protected areas, Sustainability threshold levels for Bioeconomy Technologies, Resilience of ecosystems, Energy efficiency, Soil quality, Forest quality, and Hazardous substances management.</i> - From SO 3. Dependence on non-renewable resources: <i>Bio-energy replacing non-renewable energy.</i> - From SO 5. Employment and economic competitiveness <i>Import and export of bioeconomy raw materials and products, Investments, Production and consumption of non-food and feed bio-based products, Productivity, Value added of the bioeconomy sectors.</i> The sub-topics that were found highly relevant can be found in Table 12.
4. Checking completeness sub-topics	Additional sub-topics were included linked to the identified relevant indicators from review of indicators in Tables 2 and 3. These are: - "Human health" linked to the newly proposed SO6 - "Labour rights", "Respect of indigenous rights and land rights" and "Resilience of rural communities - social protection" linked to the newly proposed SO7 - "Monitoring and accountability systems" linked to the newly proposed SO8 An additional sub-topic was suggested for SO3, namely "Supply security". Additionally, for the newly proposed SO9 Promoting global governance and collaboration a new sub-topic is considered: "partnership and co-operation".

5. Choosing a scale	For the Prato case study, it was decided to consider indicators at both the product-level as well as the regional/national level. By examining indicators at the regional/national level a broader understanding is gained of the socio-economic and environmental impacts within the region and their interactions with the Italian and European contexts. At the same time, assessing indicators at the product level allows us to delve deeper into the specific impacts and sustainability considerations associated with the textile products. Furthermore, even if the scale is classified as the product level, the indicators could be applicable to the regional level. This dual-scale approach ensures a holistic understanding of the case study.
6. Ranking of indicators	From the filtering carried out in the earlier steps, 76 indicators were left out of a total initial list of 264. This number includes product-level (66) and macro-level (10) indicators. Based on expert judgement, the indicators in the database were scored in terms of their relevance, resulting in: - 18 indicators of high relevance, - 37 indicators of medium relevance, - 21 indicators of low relevance. From the 18 indicators labelled as highly relevant, most indicators are covered by the environmental dimension (11), followed by the economic (4), and societal (2) dimension. In addition, a single highly relevant indicator was identified as a combination of dimensions: environmental and circular (1). An overview of the indicators considered of high relevance, can be found in Table 12.
7. Checking completeness indicators	28 regional level indicators were identified from the review of ISTAT analysis that are relevant for the Prato case study. These are listed in section 3.1.3, Table 4. Of these, 13 indicators were considered highly relevant at the Environmental (5), Social (4), and Economic (4) dimensions. Additionally, from the review of the LCC and s-LCA indicators (that are listed in section 3.1.2, Table 3), 10 additional (4 economic and 6 social) indicators were considered to be highly relevant. Finally, from the review of indicators from the proposed additional 3 SOs that are listed in section 3.1.1, Table 2), 1 additional social indicator was identified to be highly relevant. These additional 24 highly relevant indicators are included in Table 12 in red. Hence, in total, 42 indicators of high relevance were identified in the course of this exercise.

Table 12: Highly relevant indicators identified for the Textile sector - Recycled carded wool from Prato, Italy case study

SO	Societal Objectives	Sub-Topics	Indicators	Sustainability Dimension	Scale
2	Sustainable natural resource management	Sustainable resource use	Amount of water used in the value-chain	Environment	Product

2	Sustainable natural resource management	Sustainable resource use	Abiotic depletion (kg Sb eq.)	Environment	Product
2	Sustainable natural resource management	Sustainable resource use	Resource depletion (mineral, fossil) / product unit (MJ)	Environment	Product
2	Sustainable natural resource management	Sustainable resource use	Intensity of fossil fuel use (m3/ton of product)	Environment	Product
2	Sustainable natural resource management	Sustainable resource use	% of biological content/ product	Environment	Product
2	Sustainable natural resource management	Waste generation and management	Waste generated	Environment	Product
2	Sustainable natural resource management	Waste generation and management	% of recycled fibre raw material is used	Environment	Product
2	Sustainable natural resource management	Waste generation and management	% of materials derived from animal by-products	Environment	Product
2	Sustainable natural resource management	Sustainable resource use	Water consumption per kilogram of textiles produced in the textile industry in the region	Environment	Region
2	Sustainable natural resource management	Sustainable resource use	Percentage of recycled textile materials used in the regional textile production.	Environment	Region
2	Sustainable natural resource management	Sustainable resource use	Use of textile materials derived from sustainably managed forests (e.g. wool from sheep fed sustainable feed).	Environment	Region
2	Sustainable natural resource management	Sustainable resource use	Extent of converted or degraded agricultural land due to the expansion of the regional textile industry.	Environment	Region
2	Sustainable natural resource management	Biodiversity	Number of local threatened species protected or preserved in relation to the activities of regional textile industry.	Environment	Region
3	Dependence on non-renewable resources	Bio-material replacing non-renewable resources	Bio-based textiles	Environment	Country
3	Dependence on non-renewable resources	Material use efficiency	Material circularity indicator (MCI)	Environment, Circularity	Product

4	Mitigating and adapting to climate change	Greenhouse gas emissions	Life cycle GHG emissions (gr eq. CO ₂ /product unit)	Environment	Product
4	Mitigating and adapting to climate change	Greenhouse gas emissions	Global Warming Potential (GWP100; gr eq CO ₂)	Environment	Product
2	Sustainable natural resource management	Sustainable resource use	Residual biomass potential	Economy	Region
3	Dependence on non-renewable resources	Biomass self-sufficiency rate	Biomass from waste	Economy	Country
5	Employment and economic competitiveness	Bioeconomy-driving Policies	Strategy for creating the infrastructure and systems necessary for recovering and recycling materials (yes/no)	Economy	Product
5	Employment and economic competitiveness	Comparative advantage	Potential market share of the product	Economy	Product
5	Employment and economic competitiveness	Comparative advantage	Raw Material/Utility costs	Economy	Product
5	Employment and economic competitiveness	Employment	Personnel	Economy	Product
5	Employment and economic competitiveness	Comparative advantage	Transport	Economy	Product
5	Employment and economic competitiveness	Comparative advantage	Net Present Value	Economy	Product
5	Employment and economic competitiveness	Innovation	Investments in sustainable infrastructure for the regional textile industry (e.g. water recycling plants, waste treatment plants)	Economy	Region
7	Just inclusive and resilient economy	Resilience of rural communities - social protection	Average income of families in rural communities involved in the textile industry of the region	Economy	Region
7	Just inclusive and resilient economy	Resilience of rural communities - social protection	Financial support provided to small textile businesses in the region to promote innovation.	Economy	Region

7	Just inclusive and resilient economy	Resilience of rural communities - social protection	Investments in renewable energy technologies or infrastructure in the regional textile industry.	Economy	Region
5	Employment and economic competitiveness	Employment	Adoption of Corporate Social Responsibility (CSR) certificate (yes/no) >	Society	Product
5	Employment and economic competitiveness	Employment	Presence of working children under the legal age (yes/no)>	Society	Product
6	Disease prevention and human health	Human health	Particulate matter as respiratory inorganics / product unit (kg eq PM2.5)	Society	Product
7	Just inclusive and resilient economy	Labour rights	Forced Labour	Society	Product
7	Just inclusive and resilient economy	Labour rights	Freedom of Association	Society	Product
7	Just inclusive and resilient economy	Labour rights	Migrant Labour	Society	Product
7	Just inclusive and resilient economy	Respect of indigenous rights and land rights	Indigenous Rights	Society	Product
7	Just inclusive and resilient economy	Monitoring and accountability systems	Legal System	Society	Product
7	Just inclusive and resilient economy	Monitoring and accountability systems	Corruption	Society	Product
7	Just inclusive and resilient economy	Resilience of rural communities - social protection	Access of rural communities involved in the regional textile industry to basic services such as drinking water, education and health services.	Society	Region
7	Just inclusive and resilient economy	Resilience of rural communities - social protection	Involvement of rural communities in decisions relating to the regional textile industry.	Society	Region
9	Promoting global governance and collaboration	Partnerships and cooperation	Number of partnerships between the regional textile industry and international organisations or other global industries to promote sustainability.	Society	Region
9	Promoting global governance and collaboration	Partnerships and cooperation	Number of partnerships between the textile industry in the region and local organisations to support the development of rural communities	Society	Region

3.2.5 Textile - Man-made cellulosic fibres, like Lyocell

Man-made cellulosic fibres (MMCF) form a category of fibres produced from the cellulosic fraction of plant-based biomass. Different types of bio-based feedstocks, both primary and secondary, can be used to manufacture MMCF, with woody biomass at present having the largest share. The MMCF production chain is very complex. In short, the biomass is chipped, followed by the subsequent extraction, dissolution and regeneration of cellulose. Next, the cellulose solution is spun to fibres, extruded into filaments, washed, dried and further processed into fabrics, which can be used in a wide variety of textile applications. The stepwise consistency check carried out for this case study is presented in Table 13. The identified highly relevant indicators are listed in Table 14.

Table 13: Consistency check for the Textile - Man-made cellulosic fibres, like Lyocell case study

Methodological step	Additional methodological remarks & results
1. Filtering of societal objectives	SO 1. Food and nutrition security was considered irrelevant, as the present case study concerns non-food feedstocks (e.g., fossil oil in the reference fiber value chain, and wood, agricultural or industrial biomass from crops and residues in the MMNF value chain). There is an issue of competition with food production, but this is addressed in SO 2. Sustainable natural resource management, by the sub-topic on land use. Hence, the food and nutrition security SO was deselected. Accordingly, the following four SOs were found relevant: - SO 2. Sustainable natural resource management - SO 3. Dependence on non-renewable resources - SO 4. Mitigating and adapting to climate change - SO 5. Employment and economic competitiveness
2. Checking completeness societal objectives	The large-scale production of especially fossil-based fibers, but also MMNF, for textile purposes, are known to adversely affect the health and wellbeing of both the employees of a company, as well as the community residing in the proximity of production plants. Although the direct effects on employees may be covered by SO 5. Employment and economic competitiveness, an SO capturing the indirect societal effects is absent. Therefore, it is proposed to add another SO, concerning the health of the local community. This topic is covered by the readily proposed “Disease prevention and human health” (SO6, Table 2). Besides, the original database only partially includes indicators on labour-related aspects such as forced labour, child labour, or a beneficial working environment. Since in the global textile sector, unfavourable working conditions and the violation of labour rights tend to occur frequently, it is believed that these aspects deserve proportional attention. Therefore, it is proposed to add a SO to address these important topics, being “respect for

	labour rights” and “safe working conditions”. These are covered by the readily proposed “Just inclusive and resilient economy” SO (SO7, Table 2). Furthermore, aspects linked to good governance and compliance with the legislation (e.g., transparency in the value chain, due diligence and accountability) are essential to the complex, multi-faceted fiber production sector, and could be covered by the readily proposed “Risk assessment and monitoring” SO (SO8, Table 2). Finally, aspects linked to gender discrimination, equal pay and protection of vulnerable communities are highly relevant in the context of the labour intensive textile production sector (e.g., think of the large number of women working in textile factories). They are covered by the readily proposed “Just inclusive and resilient economy” SO (SO7, Table 2).
3. Filtering of sub-topics	Given the intricate, complex nature of the textile fiber value chains, which stretches across continents, tapping into a wide variety of resources (i.e., primary, secondary, and tertiary), as well as diverse technological operations, it carries along a multiplicity of environmental, social, and economic impacts that may not be overlooked. Each of the sub-topics in the indicators database was believed sufficiently relevant to be retained for further consideration. Hence, none of the sub-topics were deselected. The sub-topics that were found highly relevant can be found in Table 14.
4. Checking completeness sub-topics	From review of Tables 2 and 3, the following sub-topics were identified as highly relevant in light of the fiber production value chain. These are: - “Disease/hazards prevention”, under newly proposed SO6 “Disease prevention and human health”. - “Resilience of rural communities - social protection” (pertaining to topics like cultural values, religion and spiritual traditions, property rights, etc.) under newly proposed SO7 “Just inclusive and resilient economy”.
5. Choosing a scale	Both fossil-based fibers and MMNFs are semi-finished products that are part of a larger value chain, comprising different processing steps. Therefore, in the present case study it was decided to focus on indicators linked to the product-level, and to deselect the macro-level (i.e., country or regional level) indicators in the database.
6. Ranking of indicators	From the filtering carried out in the earlier steps, 157 indicators were left out of a total initial list of 264. Based on expert judgement, the indicators in the database were scored in terms of their relevance, resulting in: - 97 indicators of high relevance, - 46 indicators of medium relevance, - 6 indicators of low relevance, - 6 indicators of no relevance, - 18 indicators where the relevance is unknown (i.e., at this stage, not enough information is available for a valid judgement).

	From the 97 indicators labelled as highly relevant, most indicators are covered by the environmental dimension (77), followed by the societal (15), and economic (5) dimension. An overview of the indicators considered highly relevant, can be found in Table 14.
7. Checking completeness indicators	No additional indicators were identified.

Table 14: Highly relevant indicators identified for the Textile sector - Man-made cellulosic fibres, like Lyocell case study

	Societal objectives	Sub-topics	Indicators	Domains
2	Sustainable natural resource management	Biodiversity	Ha of protected areas and land with significant biodiversity values used / product unit	Environment
2	Sustainable natural resource management	Biodiversity	% of biomass produced on land that is/was highly biodiverse grassland	Environment
2	Sustainable natural resource management	Biodiversity	% of biomass produced on land that is/was a protected area	Environment
2	Sustainable natural resource management	Biodiversity	Functional diversity (Number and variety of the elements of biodiversity that influence how ecosystems function)	Environment
2	Sustainable natural resource management	Biodiversity	Biodiversity damage potential (BDP; m2*year*PAS (potentially affected species)	Environment
2	Sustainable natural resource management	Land use change (LUC)	% of biomass obtained from land with high carbon stock (e.g. wetlands)	Environment
2	Sustainable natural resource management	Land use change (LUC)	% of biomass obtained from land that is/was peatland	Environment
2	Sustainable natural resource management	Land use change (LUC)	% of crop supply, by mass, obtained from land that is/was primary forest or other wooded land	Environment
2	Sustainable natural resource management	Land use change (LUC)	Agricultural land converted to energy (or bio-based product) crop	Environment

2	Sustainable natural resource management	Land use change (LUC)	% of biofuels and bioliquids made from raw material obtained from land that is/was continuously forested	Environment
2	Sustainable natural resource management	Land use change (LUC)	Change in wood/wood fibre demand for forest products	Environment
2	Sustainable natural resource management	Land use change (LUC)	Grassland converted to cropland	Environment
2	Sustainable natural resource management	Land use change (LUC)	Forest converted to cropland	Environment
2	Sustainable natural resource management	Primary Biomass production	Wood consumption	Environment
2	Sustainable natural resource management	Sustainable resource use	Amount of water used in the value-chain	Environment
2	Sustainable natural resource management	Sustainable resource use	Estimated amount of water used for irrigation (m3)	Environment
2	Sustainable natural resource management	Sustainable resource use	Water exploitation index	Environment
2	Sustainable natural resource management	Sustainable resource use	% of irrigated crops and freshwater intensive operation systems established in long-term freshwater-stressed areas	Environment
2	Sustainable natural resource management	Sustainable resource use	Water consumption (use) (m3/product unit processed)	Environment
2	Sustainable natural resource management	Sustainable resource use	Amount of organic or mineral fertilizers and pesticides used (kg)	Environment
2	Sustainable natural resource management	Sustainable resource use	Abiotic depletion (kg Sb eq.)	Environment
2	Sustainable natural resource management	Sustainable resource use	Resource depletion (mineral, fossil) / product unit (MJ)	Environment
2	Sustainable natural resource management	Sustainable resource use	Intensity of fossil fuel use (m3/ton of product)	Environment

2	Sustainable natural resource management	Sustainable resource use	Change in consumption level of fossil fuel resources / product unit	Environment
2	Sustainable natural resource management	Sustainable resource use	% of biological content/ product	Environment
2	Sustainable natural resource management	Sustainable resource use	% of carbon content derived from renewable raw materials	Environment
2	Sustainable natural resource management	Energy efficiency	Renewable energies (RES) share in final energy consumption / product unit	Environment
2	Sustainable natural resource management	Energy efficiency	Renewable electricity used for manufactured products (kWh or % of total electricity used)	Environment
2	Sustainable natural resource management	Energy efficiency	Primary energy savings (kWh or % of total primary energy consumed)	Environment
2	Sustainable natural resource management	Energy efficiency	Energy Intensity (product value added / mJ primary energy)	Environment
2	Sustainable natural resource management	Energy efficiency	Energy consumption by activity/ production process/unit of production (kWh/production process or product unit)	Environment
2	Sustainable natural resource management	Water quality	Ecotoxicity for aquatic fresh water (Comparative Toxic Unit for ecosystems)	Environment
2	Sustainable natural resource management	Water quality	Volume of water leaving the manufacturing facility meets drinking water quality standards (m3 or as percentage of total amount of water leaving the manufacturing facility)	Environment
2	Sustainable natural resource management	Forest quality	% of crop supply, by mass, was grown on fields with zero conversion High Carbon Stock (HCS) forests	Environment
2	Sustainable natural resource management	Forest quality	% of crop supply, by mass, was grown on fields with zero conversion of High Conservation Value (HCV) forests	Environment
2	Sustainable natural resource management	Forest quality	% of the total amount of wood used comes from sustainable forest management (economically viable, environmentally sound, socially responsible) or are waste wood according to waste wood categories	Environment
2	Sustainable natural resource management	Forest quality	Share of forests certified for sustainable management	Environment

2	Sustainable natural resource management	Forest quality	% of deforestation or afforestation /product unit	Environment
2	Sustainable natural resource management	Hazardous substances management	% of plastic (synthetic plastic material, plasticizers, PVC) contained in the product	Environment
2	Sustainable natural resource management	Hazardous substances management	% of hazardous colorants	Environment
2	Sustainable natural resource management	Hazardous substances management	% of chlorine, halogenated bleaching or directly biodegradable complexing agents	Environment
2	Sustainable natural resource management	Hazardous substances management	Hazardous chemicals use reduction	Environment
2	Sustainable natural resource management	Hazardous substances management	% of product materials produced and managed to high environmental and social standards	Environment
2	Sustainable natural resource management	Waste generation and management	Waste generated	Environment
2	Sustainable natural resource management	Waste generation and management	Procedures for separating and using recyclable materials from the waste stream (yes/no)	Environment
2	Sustainable natural resource management	Waste generation and management	Presence of an irrigation and water distribution system that minimize water waste (yes/no)	Environment
2	Sustainable natural resource management	Waste generation and management	Declaration of end-of-life disposal (in the CSR report) (yes/no)	Environment
2	Sustainable natural resource management	Waste generation and management	Procedures for recovering materials for other uses, such as incineration for raising process steam or heating, or agricultural use (yes/no)	Environment
2	Sustainable natural resource management	Waste generation and management	% of recycled fibre raw material is used	Environment
2	Sustainable natural resource management	Waste generation and management	% of product is actively being recovered and recycled	Environment
2	Sustainable natural resource management	Waste generation and management	% of materials derived from recycling or recovery of any other biomass waste	Environment

2	Sustainable natural resource management	Waste generation and management	% of materials derived from any other biomass by-products	Environment
2	Sustainable natural resource management	Waste generation and management	% of materials derived from fecal matter, straw and other natural non-hazardous agricultural or forestry material	Environment
2	Sustainable natural resource management	Waste generation and management	Practices to segregate different waste types to facilitate reuse, recycling or composting (yes/no)	Environment
2	Sustainable natural resource management	Waste generation and management	% of the total amount of wood used is waste wood according to waste wood categories	Environment
2	Sustainable natural resource management	Waste generation and management	% of the total amount of agricultural biomass used come from agricultural residues	Environment
2	Sustainable natural resource management	Waste generation and management	% of reused and/or recycled wastewater	Environment
2	Sustainable natural resource management	Waste generation and management	% of by-products or wastes reused by the processing/production facility or transferred to other sectors	Environment
2	Sustainable natural resource management	Waste generation and management	Recycling rate/cascading indexes	Environment
2	Sustainable natural resource management	Waste generation and management	% of product-related process chemicals in effluent are directly discharged into a water body	Environment
2	Sustainable natural resource management	Waste generation and management	% of product-related chemicals are contained in effluent	Environment
2	Sustainable natural resource management	Waste generation and management	Amount of wastewater from processing operations is discharged into aquatic systems (m3)	Environment
2	Sustainable natural resource management	Waste generation and management	Amount of sewage is discharged into aquatic systems (m3)	Environment
2	Sustainable natural resource management	Waste generation and management	% of biodegradability	Environment
2	Sustainable natural resource management	Waste generation and management	Procedures for handling, collecting, separating and disposal of hazardous waste as defined by the relevant local and national regulatory authorities (yes/no)	Environment

2	Sustainable natural resource management	Waste generation and management	Practices for waste storage, treatment and disposal that do not pose health or safety risks to farmers, workers, other people, or natural ecosystems (yes/no)	Environment
2	Sustainable natural resource management	Waste generation and management	% of chemical waste and empty containers is collected and disposed in compliance with good practices	Environment
2	Sustainable natural resource management	Waste generation and management	% of plastic fibres	Environment
2	Sustainable natural resource management	Waste generation and management	Post-consuming recycling rate/cascading indexes	Environment
3	Dependence on non-renewable resources	Bio-energy replacing non-renewable energy	% of biofuels produced with co-products, residues and waste	Environment
3	Dependence on non-renewable resources	Material use efficiency	Carbon intensity	Environment
3	Dependence on non-renewable resources	Material use efficiency	Biomass Utilisation Factor (BUF)	Environment
4	Mitigating and adapting to climate change	Greenhouse gas emissions	Life cycle GHG emissions (gr eq. CO ₂ /product unit)	Environment
4	Mitigating and adapting to climate change	Greenhouse gas emissions	Rate of GHG emissions reduction or savings	Environment
4	Mitigating and adapting to climate change	Greenhouse gas emissions	Potential leakages (gr eq. CO ₂ / product unit)	Environment
4	Mitigating and adapting to climate change	Greenhouse gas emissions	Global Warming Potential (GWP100; gr eq CO ₂)	Environment
4	Mitigating and adapting to climate change	Air quality	Climate regulation potential (CRP) (ton C/m ³)	Environment
5	Employment and economic competitiveness	Bioeconomy-driving Policies	Strategy for creating the infrastructure and systems necessary for recovering and recycling materials (yes/no)	Economy
5	Employment and economic competitiveness	Comparative advantage	Potential market share of the product	Economy

5	Employment and economic competitiveness	Employment	Full time equivalent jobs along the full value chain of a product	Economy
5	Employment and economic competitiveness	Innovation	Number of patents on resource efficiency technologies	Economy
5	Employment and economic competitiveness	Value Added of the bioeconomy sectors	Contribution of the product/service/organization to economic progress (revenue, gain, paid wages, R+D costs in relation to revenue, etc.)	Economy
5	Employment and economic competitiveness	Bioeconomy-driving Policies	Presence of formal policies on equal opportunities	Society
5	Employment and economic competitiveness	Bioeconomy-driving Policies	Presence of a formal policy in place concerning health and safety of all workers, including contractors (yes/no)	Society
5	Employment and economic competitiveness	Bioeconomy-driving Policies	Existence of (legal) obligation on public sustainability report (yes/no)	Society
5	Employment and economic competitiveness	Stakeholder engagement and capacity building	Presence of a law or norm regarding transparency (yes/no)	Society
5	Employment and economic competitiveness	Stakeholder engagement and capacity building	Non-compliance with regulations regarding transparency (yes/no)	Society
5	Employment and economic competitiveness	Innovation	Number of certifications/labels the organization obtained for the product/site	Society
5	Employment and economic competitiveness	Employment	% of crop supply came from human rights low-risk countries with corrective actions taken for any known high-risk sites	Society
5	Employment and economic competitiveness	Employment	% of crop supply came from human rights high-risk countries that have high-risk sites for which we took corrective actions	Society
5	Employment and economic competitiveness	Employment	Risk of forced labour used for production of commodity (yes/no)	Society
5	Employment and economic competitiveness	Employment	Evidence of restriction to freedom of association and collective bargaining (yes/no)	Society
5	Employment and economic competitiveness	Employment	Adoption of Corporate Social Responsibility (CSR) certificate (yes/no) [	Society

5	Employment and economic competitiveness	Employment	Presence of working children under the legal age (yes/no)	Society
5	Employment and economic competitiveness	Employment	Change in unpaid time spent by women and children collecting biomass /product unit produced	Society
5	Employment and economic competitiveness	Innovation	Presence of environmentally/friendly technologies (yes/no)	Society
5	Employment and economic competitiveness	Productivity	Land for biomass production	Society

3.2.6 Textile - Latxa sheep wool from the Basque Country of Spain

This case study is geared towards the valorisation of conventionally discarded wool from Latxa sheep in the Basque Country, situated in Spain. For years, Latxa sheep wool was typically incinerated, instead of being regarded as a valuable feedstock. Recently, TERNUA, a Basque apparel manufacturer, has designed a new jacket using Latxa wool as insulation material. Henceforth, the conventionally discarded sheep wool can be kept in the loop and contribute to the circularity of the sportswear sector. Simultaneously, this regional production system results in a reduction of both regional waste volumes and the import of carbon-intensive insulation materials. The stepwise consistency check carried out for this case study is presented in Table 15. The identified highly relevant indicators are listed in Table 16.

Table 15: Consistency check for the Textile - Latxa sheep wool from the Basque Country of Spain case study

Methodological step	Additional methodological remarks & results
1. Filtering of societal objectives	SO 1. Food nutrition and security was considered irrelevant, as the feedstock of interest in the present case study concerns waste (i.e., wool), which does not compete with food-related applications. Hence, this SO was deselected. Accordingly, these four SOs were found relevant in this case study: - SO 2. Sustainable natural resource management - SO 3. Dependence on non-renewable resources - SO 4. Mitigating and adapting to climate change - SO 5. Employment and economic competitiveness
2. Checking completeness societal objectives	The selected SOs sufficiently cover the environmental pillar, and, to a great extent, also the economic pillar. However, a major gap was identified for the societal pillar, as SO 5. Employment and economic competitiveness does not in detail address the health and wellbeing of people, including the economic resilience of rural communities. Along these lines, as textile value chains, including TERNUA's, are generally global, exceptional

	relevance is given to specific categories of stakeholders, such as local communities (i.e., farmers) and manufacturers, often located in developing countries. Hence, since SO 5 seems more oriented to innovation and economic competitiveness, it is believed that an additional SO focused on “Human health and wellbeing” is needed. This is covered within the already proposed SO6 Disease prevention and human health.
3. Filtering of sub-topics	Due to the nature of the TERNUA case study, which focuses on repurposing currently discarded wool sheep as insulating material, the following subtopics were considered not relevant and hence excluded. - SO2. Sustainable natural resource management: <i>Biodiversity, Land cover/use, Indirect Land use change (iLUC), Land use change (LUC), Biomass production, Sustainability threshold levels for Bioeconomy Technologies, Resilience of ecosystems, Forest quality, and Hazardous substances management.</i> - SO3. Dependence on non-renewable resources: <i>Bio-energy replacing non-renewable energy, Biomass self-sufficiency rate</i> - SO4. Mitigating and adapting to climate change: <i>Climate change adaptation</i> The sub-topics that were found highly relevant can be found in Table 16.
4. Checking completeness sub-topics	Additional sub-topics were included linked to the identified relevant indicators from review of indicators in Tables 2 and 3. These are: - “Human health” linked to the newly proposed SO6 - “Labour rights” and “Respect of indigenous rights and land rights” linked to the newly proposed SO7 - “Monitoring and accountability systems” linked to the newly proposed SO8 This inclusion was based on the additional indicators that were found relevant to include in the TERNUA case study .
5. Choosing a scale	The TERNUA case study was formerly conducted at product-level within the ORIENTING project. The objective of the study was to carry out an evaluation of the life cycle sustainability of a jacket. A principal component of the jacket was the use of wool (as insulation material) generally discarded as waste. Last year, the TERNUA company launched this new product on the market with a limited edition. Since this business model is based on commitment to the local herding community, who benefit from the sale of the wool and will save on incineration costs, simultaneously inducing substantial environmental improvements, it was decided to explore the upscaling of the case study considering territorial impacts - mostly at local and regional level (i.e., Basque Country, Spain).
6. Ranking of indicators	From the filtering carried out in the earlier steps, 54 indicators were selected out of 264, all at the product-level.

	Based on expert judgement, the indicators in the database were scored in terms of their relevance, resulting in: - 12 indicators of high relevance, - 25 of medium relevance, - 17 of low relevance. The UN report “Sustainability and Circularity in the Textile Value Chain” was reviewed to support the prioritization analysis. All of the 12 indicators labelled as highly relevant, are covered by the environmental dimension. An overview of the indicators considered highly relevant, can be found in Table 16.
7. Checking completeness indicators	For the TERNUA case study, the additional product-level indicators proposed in section 3.1, Table 2 and 3 were reviewed. From these indicators, 11 were classified as highly relevant. They are included in Table 16 in red. Additionally, for the regional level, the indicators proposed in section 3.1.3, Table 4 were reviewed, and 15 were found to be relevant. Hence, in total, 38 indicators of high relevance were identified in the course of this exercise.

Table 16: Highly relevant indicators identified for the Textile sector - Latxa sheep wool from the Basque Country of Spain case study

SO	Societal objectives	Sub-topics	Indicators	Domains	Scale
2	Sustainable natural resource management	Sustainable resource use	% of biological content/ product	Environment	Product
2	Sustainable natural resource management	Energy efficiency	Energy Intensity (product value added / mJ primary energy)	Environment	Product
2	Sustainable natural resource management	Energy efficiency	Energy consumption by activity/ production process/unit of production (kWh/production process or product unit)	Environment	Product
2	Sustainable natural resource management	Waste generation and management	Waste generated	Environment	Product
2	Sustainable natural resource management	Waste generation and management	Procedures for separating and using recyclable materials from the waste stream (yes/no)	Environment	Product
2	Sustainable natural resource management	Waste generation and management	% of recycled materials for packaging	Environment	Product

2	Sustainable natural resource management	Waste generation and management	% of materials derived from animal by-products	Environment	Product
2	Sustainable natural resource management	Sustainable resource use	Resource depletion (mineral, fossil) / product unit (MJ)	Environment	Product
2	Sustainable natural resource management	Sustainable resource use	Intensity of fossil fuel use (m3/ton of product)	Environment	Product
3	Dependence on non-renewable resources	Material use efficiency	Material circularity indicator (MCI)	Environment, Circularity	Product
4	Mitigating and adapting to climate change	Greenhouse gas emissions	Global Warming Potential (GWP100; gr eq CO2)	Environment	Product
4	Mitigating and adapting to climate change	Greenhouse gas emissions	Life cycle GHG emissions (gr eq. CO2/product unit)	Environment	Product
5	Employment and economic competitiveness	Comparative advantage	Raw Material/Utility costs	Economy	Product
5	Employment and economic competitiveness	Employment	Personnel	Economy	Product
5	Employment and economic competitiveness	Comparative advantage	Transport	Economy	Product
5	Employment and economic competitiveness	Comparative advantage	Life cycle cost (or total undiscounted cost)	Economy	Product

6	Disease prevention and human health	Human health	Particulate matter as respiratory inorganics / product unit (kg eq PM2.5)	Society	Product
7	Just inclusive and resilient economy	Labour rights	Forced Labour	Society	Product
7	Just inclusive and resilient economy	Labour rights	Freedom of Association	Society	Product
7	Just inclusive and resilient economy	Labour rights	Migrant Labour	Society	Product
7	Just inclusive and resilient economy	Respect of indigenous rights and land rights	Indigenous Rights	Society	Product
8	Risk assessment and monitoring	Monitoring and accountability systems	Legal System	Society	Product
8	Risk assessment and monitoring	Monitoring and accountability systems	Corruption	Society	Product
2	Sustainable natural resource management	Waste generation and management	Percentage of biological materials (such as wool) used in the textile industry in the region compared to total materials used.	Environment	Region
2	Sustainable natural resource management	Sustainable resource use	Water consumption per kilogram of textiles produced in the textile industry in the region	Environment	Region
2	Sustainable natural resource management	Sustainable resource use	Material extracted compared to economic value added (kg/€), OR reduction of CO2 emissions per unit of textile production compared to previous years.	Environment	Region
2	Sustainable natural resource management	Sustainable resource use	Percentage of recycled textile materials used in the regional textile production.	Environment	Region
2	Sustainable natural resource management	Sustainable resource use	Use of textile materials derived from sustainably managed forests (e.g. wool from sheep fed sustainable feed).	Environment	Region
5	Employment and economic competitiveness	Innovation	N° of new sustainable technologies integrated into the regional textile industry	Economy	Region
5	Employment and economic competitiveness	Innovation	Research and development investments in the textile industry in the region (in relation to total turnover)	Economy	Region
5	Employment and economic competitiveness	Innovation	Investments in sustainable infrastructure for the regional textile industry (e.g. water recycling plants, waste treatment plants)	Economy	Region
7	Just inclusive and resilient economy	Resilience of rural communities - social protection	Average income of families in rural communities involved in the textile industry of the region	Economy	Region

7	Just inclusive and resilient economy	Resilience of rural communities - social protection	Financial support provided to small textile businesses in the region to promote innovation.	Economy	Region
7	Just inclusive and resilient economy	Monitoring and accountability systems	Quantity of textile waste generated compared to total production in the regional textile industry.	Economy	Region
7	Just inclusive and resilient economy	Resilience of rural communities - social protection	Involvement of rural communities in decisions relating to the regional textile industry.	Society	Region
7	Just inclusive and resilient economy	Inclusiveness	Number of training programmes or technical cooperation of the regional textile industry with developing countries to promote sustainability.	Society	Region
9	Promoting global governance and collaboration	Partnerships and cooperation	Number of partnerships between the regional textile industry and international organisations or other global industries to promote sustainability.	Society	Region
9	Promoting global governance and collaboration	Partnerships and cooperation	Number of partnerships between the textile industry in the region and local organisations to support the development of rural communities	Society	Region

3.2.7 Chemical - Chemicals from waste

This value chain concerns the conversion of (non-recyclable, non-compostable) municipal solid waste into methanol. It is produced via the gasification of municipal solid waste into syngas. The produced methanol is a drop-in chemical, chemically identical to fossil-based methanol that can be used to produce a variety of everyday products, such as clothing, adhesives, paint, pharmaceuticals, building materials and car parts. The stepwise consistency check carried out for this case study is presented in Table 17. The identified highly relevant indicators are listed in Table 18.

Table 17: Consistency check for the Chemical - Chemicals from waste case study

Methodological step	Additional methodological remarks & results
1. Filtering of societal objectives	SO 1. Food and nutrition security was considered irrelevant, as the feedstock of interest in the present case study concerns waste, which does not compete with food-related applications. Hence, this SO was deselected. Accordingly, four SOs were found relevant in this case study: - SO 2. Sustainable natural resource management - SO 3. Dependence on non-renewable resources - SO 4. Mitigating and adapting to climate change - SO 5. Employment and economic competitiveness

2. Checking completeness societal objectives	The processing of waste and large-scale production of chemicals (including methanol) may affect the health and wellbeing of both the employees of a company, as well as the local community, and the general public. Although the direct effects on employees may be covered by SO 5. Employment and economic competitiveness, a SO capturing the indirect societal effects is absent. Therefore, it is proposed to add another SO related to Health and wellbeing of people and another related to gender equality and fair salary. These additional aspects are covered within the already proposed SO6 “Disease prevention and human health” and SO7 “Just inclusive and resilient economy” in Table 2.
3. Filtering of sub-topics	As the present case study does not directly involve any biomass cultivation, processing or trading activities, a set of sub-categories linked to agro-industrial production systems were excluded. These include the following: - From SO 2. Sustainable natural resource management: <i>Biodiversity, Indirect Land use change (iLUC), Land cover/use, Land use change (LUC), Primary biomass production, Protected areas, Resilience of ecosystems, Soil quality, Forest quality and Marine environments’ quality.</i> - From SO 3. Dependence on non-renewable resources: <i>Biomass self-sufficiency rate, Bio-material replacing non-renewable resources, and Certified bio-based products.</i> - From SO 5. Employment and economic competitiveness: <i>Import and export of bioeconomy raw materials and products.</i> The sub-topics that were found highly relevant can be found in Table 18.
4. Checking completeness sub-topics	As a sub-topic specifically linked to the economic prosperity or profitability of a production system, and its effect on human health, gender equality and working conditions, appear absent in the database, four new sub-topics are proposed: - “Economic viability”, to SO5 “Employment and economic competitiveness” besides the sub-topic proposed in Table 3: “Externalities” - “Human health” linked to the newly proposed SO6. - “Gender equality” and “Labour rights” linked to the newly proposed SO7. This inclusion was based on the additional indicators that were found relevant, as well as the sub-topics these are linked to.
5. Choosing a scale	The waste-to-methanol case study links to a specific value chain, covering the conversion from municipal solid waste (MSW) to methanol as a platform chemical. Therefore, it was decided to adopt the indicators linked to the product-level, and to deselect the macro-level (i.e., country or regional level) focused indicators in the database.
6. Ranking of indicators	From the filtering carried out in the earlier steps, 139 indicators were selected out of a total initial list of 264. A literature review was performed, specifically targeting the environmental, economic and social sustainability aspects of the

	waste-to-methanol value chains. Several search queries were conducted, combining the following key words: <i>“social”, “economic”, “environmental”, “life cycle assessment”, “life cycle costing”, “impacts”, “methanol”, “chemicals”, “production”, “value chain”, “waste”, “municipal solid waste”</i> Based on this search, the indicators in the database were scored in terms of their relevance, resulting in: - 19 indicators of high relevance, - 15 indicators of low relevance, - 49 indicators of no relevance, - 50 indicators where the relevance is unknown. From the 19 indicators labelled as highly relevant, most indicators are covered by the environmental dimension (14), followed by the societal (3), and economic (2) dimension. An overview of the indicators considered highly relevant, can be found in Table 18.
7. Checking completeness indicators	During the literature review, 12 indicators were identified to be highly relevant that are not yet covered in the original database (Iaquaniello et al., 2017, Iribarren et al., 2022). These include economic (8), as well as social (3) and environmental (1) indicators. They are included in Table 18 in red. Each of these indicators has been used to measure the performance of the waste-to-methanol value chain in literature studies, compared to the reference methanol value chain, and was therefore considered of high relevance. Hence, in total, 31 indicators of high relevance were identified in the course of this exercise.

Table 18: Highly relevant indicators identified for the Chemical - Chemicals from waste case study

SO	Societal objectives	Sub-topics	Indicators	Domains
2	Sustainable natural resource management	Sustainable resource use	Water consumption (use) (m3/product unit processed)	Environment
2	Sustainable natural resource management	Energy efficiency	Renewable electricity used for manufactured products (kWh or % of total electricity used)	Environment
2	Sustainable natural resource management	Energy efficiency	Energy consumption by activity/ production process/unit of production (kWh/production process or product unit)	Environment

2	Sustainable natural resource management	Waste generation and management	Waste generated	Environment
2	Sustainable natural resource management	Waste generation and management	Procedures for separating and using recyclable materials from the waste stream (yes/no)	Environment
2	Sustainable natural resource management	Waste generation and management	Procedures for recovering materials for other uses, such as incineration for raising process steam or heating, or agricultural use (yes/no)	Environment
2	Sustainable natural resource management	Waste generation and management	% of materials totally or partially derived from the organic fraction of mixed municipal household waste separated through mechanical, physicochemical, biological and/or manual treatment	Environment
2	Sustainable natural resource management	Waste generation and management	Practices to segregate different waste types to facilitate reuse, recycling or composting (yes/no)	Environment
2	Sustainable natural resource management	Waste generation and management	% of by-products or wastes reused by the processing/production facility or transferred to other sectors	Environment
2	Sustainable natural resource management	Waste generation and management	Practices for waste storage, treatment and disposal that do not pose health or safety risks to farmers, workers, other people, or natural ecosystems (yes/no)	Environment
3	Dependence on non-renewable resources	Bio-energy replacing non-renewable energy	% of biofuels produced with co-products, residues and waste	Environment
3	Dependence on non-renewable resources	Material use efficiency	Carbon intensity	Environment
4	Mitigating and adapting to climate change	Greenhouse gas emissions	Global Warming Potential (GWP100; gr eq CO2)	Environment
4	Mitigating and adapting to climate change	Air quality	Emissions of non-GHG air pollutants (including air toxics) (µg / product unit)	Environment
4	Mitigating and adapting to climate change	Air quality	Ozone depletion (mg CFC-11 eq/kg)	Environment

5	Employment and economic competitiveness	Bioeconomy-driving Policies	Strategy for creating the infrastructure and systems necessary for recovering and recycling materials (yes/no)	Economy
5	Employment and economic competitiveness	Comparative advantage	Potential market share of the product	Economy
5	Employment and economic competitiveness	Economic viability	Cost-competitiveness with the conventional, incumbent product	Economy
5	Employment and economic competitiveness	Economic viability	Payback time of the investment	Economy
5	Employment and economic competitiveness	Economic viability	Return on investment (ROI)	Economy
5	Employment and economic competitiveness	Economic viability	Cost of production (COP)	Economy
5	Employment and economic competitiveness	Externalities	Internalised external costs	Economy
5	Employment and economic competitiveness	Comparative advantage	Capital costs	Economy
5	Employment and economic competitiveness	Comparative advantage	Operating costs	Economy
5	Employment and economic competitiveness	Comparative advantage	Life cycle cost (or total undiscounted cost)	Economy
5	Employment and economic competitiveness	Stakeholder engagement and capacity building	Number and variety of relevant stakeholders participating in the consultative process	Society
5	Employment and economic competitiveness	Employment	Adoption of Corporate Social Responsibility (CSR) certificate (yes/no)	Society
5	Employment and economic competitiveness	Innovation	Involvement in technology transfer program or projects (yes/no)	Society
6	Disease prevention and human health	Human health	Health expenditure	Society

7	Just and inclusive resilient economy	Gender equality	Women in the labour force	Society
7	Just and inclusive resilient economy	Labour rights	Fair salary	Society

3.2.8 Chemical - Bio-MEG, bio-MPG, and renewable functional fillers from wood

Monoethylene Glycol (MEG) is an organic compound, produced from the industrial hydrolysis of ethylene oxide. It has various fields of application, such as the production of polyester fibres and polyethylene terephthalate resins, as well as antifreeze in coolant agents, which are used in a broad range of manufacturing industries. MEG produced from agricultural residues and lignocellulose-rich raw materials, mostly known as bio-MEG, offers the possibility of a reduction in GHG emissions and decreased dependency on finite resources compared to its fossil-based counterpart. The stepwise consistency check carried out for this case study is presented in Table 19. The identified highly relevant indicators are listed in Table 20.

Table 19: Consistency check for the Chemical - Bio-MEG, bio-MPG, and renewable functional fillers from wood case study

Methodological step	Additional methodological remarks & results
1. Filtering of societal objectives	SO 1. Food and nutrition security was considered as not relevant for this case, as the feedstock of interest in the present case study is wood and wood residues, which does not compete with food-related applications. Hence, this SO was deselected in the Excel filtering functionality. The production of BioMEG and BioMPG could also result from sugars derived from agriculture residues. In case we would consider a wider feedstock source in the production of these platform chemicals, the filtering out of SO.1 should be reconsidered. The following four SOs were found relevant for this case study: 2. Sustainable natural resource management 3. Dependence on non-renewable resources 4. Mitigating and adapting to climate change 5. Employment and economic competitiveness
2. Checking completeness societal objectives	No additional SOs were identified.
3. Filtering of sub-topics	The following sub-topics were filtered out due to not being relevant for this case study: - From SO 2. Sustainable natural resource management: <i>Sustainability threshold levels for Bioeconomy Technologies, Marine</i>

	<i>environments' quality, and Resilience of ecosystems.</i> The second sub-topic related to marine was found outside of scope. As for the third sub-topic, this is left out as the indicators are very specific to the capacity of ecosystem services. The sub-topics that were found highly relevant can be found in Table 20.
4. Checking completeness sub-topics	No additional sub-topics were identified.
5. Choosing a scale	Wood resources and residues to BioMEG and BioMPG refers to a value chain scale, covering the collection of feedstocks from diverse sources (e.g., sustainable forest as well as sawmill residues), feedstock treatment, and conversion towards the platform chemicals. Therefore, it was decided to adopt the indicators linked to the product-level, and to deselect the macro level indicators in the database.
6. Ranking of indicators	The production of BioMEG and BioMPG platform chemicals from strictly wood resources is a relatively new subject and therefore only few resources were found to consult about environmental, economic and societal sustainability aspects of this value chain. The literature review performed consisted of identified relevant scientific papers and other reports describing diverse processes for the production of BioMEG and BioMPG from biomass sources. As well as other studies in relation to sustainability assessment of forest resources use (harvesting, transport) to complement the identified information gaps. After the filtering exercise a total of 164 indicators were still available. Their relevance for the case study was scored based on the literature research as follows: - 43 indicators of high relevance, - 77 indicators of low relevance, - 35 indicators of no relevance, - 9 indicators where the relevance is unknown. An overview of the indicators considered highly relevant, can be found in Table 20.
7. Checking completeness indicators	No additional sub-topics were identified.

Table 20: Highly relevant indicators identified for the Chemical - Bio-MEG, bio-MPG, and renewable functional fillers from wood case study

SO	Societal objectives	Sub-topics	Indicators	Domains
2	Sustainable natural resource management	Biodiversity	Ha of protected areas and land with significant biodiversity values used / product unit	Environment
2	Sustainable natural resource management	Biodiversity	Biodiversity damage potential (BDP; m ² *year*PAS (potentially affected species)	Environment
2	Sustainable natural resource management	Land use change (LUC)	Change in wood/wood fibre demand for forest products	Environment
2	Sustainable natural resource management	Primary Biomass production	Wood consumption	Environment
2	Sustainable natural resource management	Sustainable resource use	Amount of water used in the value-chain	Environment
2	Sustainable natural resource management	Sustainable resource use	Water consumption (use) (m ³ /product unit processed)	Environment
2	Sustainable natural resource management	Sustainable resource use	Intensity of fossil fuel use (m ³ /ton of product)	Environment
2	Sustainable natural resource management	Sustainable resource use	% of biological content/ product	Environment
2	Sustainable natural resource management	Sustainable resource use	% of carbon content derived from renewable raw materials	Environment
2	Sustainable natural resource management	Energy efficiency	Renewable energies (RES) share in final energy consumption / product unit	Environment
2	Sustainable natural resource management	Energy efficiency	Energy consumption by activity/ production process/unit of production (kWh/production process or product unit)	Environment
2	Sustainable natural resource management	Water quality	Ecotoxicity for aquatic fresh water (Comparative Toxic Unit for ecosystems)	Environment
2	Sustainable natural resource management	Water quality	Acidification (mol H ⁺ eq from NO _x , SO _x , NH ₃)	Environment
2	Sustainable natural resource management	Water quality	Eutrophication (gr eq PO ₄)	Environment
2	Sustainable natural resource management	Soil quality	Soil erosion / product unit	Environment
2	Sustainable natural resource management	Soil quality	Soil nutrient balance	Environment

2	Sustainable natural resource management	Forest quality	% of the total amount of wood used comes from sustainable forest management (economically viable, environmentally sound, socially responsible) or are waste wood according to waste wood categories	Environment
2	Sustainable natural resource management	Forest quality	% of deforestation or afforestation /product unit	Environment
2	Sustainable natural resource management	Hazardous substances management	% of hazardous (carcinogenic, mutagenic, reprotoxic) substances	Environment
2	Sustainable natural resource management	Hazardous substances management	% of product materials produced and managed to high environmental and social standards	Environment
2	Sustainable natural resource management	Waste generation and management	Waste generated	Environment
2	Sustainable natural resource management	Waste generation and management	Procedures for separating and using recyclable materials from the waste stream (yes/no)	Environment
2	Sustainable natural resource management	Waste generation and management	Declaration of end-of-life disposal (in the CSR report) (yes/no)	Environment
2	Sustainable natural resource management	Waste generation and management	% of materials derived from recycling or recovery of any other biomass waste	Environment
2	Sustainable natural resource management	Waste generation and management	Recycling rate/cascading indexes	Environment
2	Sustainable natural resource management	Waste generation and management	Procedures for handling, collecting, separating and disposal of hazardous waste as defined by the relevant local and national regulatory authorities (yes/no)	Environment
2	Sustainable natural resource management	Waste generation and management	Practices for waste storage, treatment and disposal that do not pose health or safety risks to farmers, workers, other people, or natural ecosystems (yes/no)	Environment
2	Sustainable natural resource management	Waste generation and management	% of chemical waste and empty containers is collected and disposed in compliance with good practices	Environment
2	Sustainable natural resource management	Waste generation and management	Post-consuming recycling rate/cascading indexes	Environment
3	Dependence on non-renewable resources	Material use efficiency	Biomass Utilisation Factor (BUF)	Environment , Circularity
4	Mitigating and adapting to climate change	Greenhouse gas emissions	Life cycle GHG emissions (gr eq. CO ₂ /product unit)	Environment
4	Mitigating and adapting to climate change	Greenhouse gas emissions	Global Warming Potential (GWP100; gr eq CO ₂)	Environment
4	Mitigating and adapting to climate change	Air quality	Amount of volatile organic compounds (VOC) emitted (parts per billion (ppb), parts per million (ppm), or as micrograms per cubic meter (µg/m ³))	Environment

5	Employment and economic competitiveness	Innovation	Presence of environmentally/friendly technologies (yes/no)	Environment
5	Employment and economic competitiveness	Bioeconomy-driving Policies	Strategy for creating the infrastructure and systems necessary for recovering and recycling materials (yes/no)	Economy
5	Employment and economic competitiveness	Comparative advantage	Potential market share of the product	Economy
5	Employment and economic competitiveness	Value Added of the bioeconomy sectors	Contribution of the product/service/organization to economic progress (revenue, gain, paid wages, R+D costs in relation to revenue, etc.)	Economy
5	Employment and economic competitiveness	Employment	Full time equivalent jobs along the full value chain of a product	Economy, Society
5	Employment and economic competitiveness	Bioeconomy-driving Policies	Presence of a formal policy in place concerning health and safety of all workers, including contractors (yes/no)	Society
5	Employment and economic competitiveness	Stakeholder engagement and capacity building	Number of informal workshops to build local understanding in the community of the processes that may impact them directly to aid meaningful engagement	Society
5	Employment and economic competitiveness	Stakeholder engagement and capacity building	Promoting the involvement of small holders or small suppliers (yes/no)	Society
5	Employment and economic competitiveness	Stakeholder engagement and capacity building	Presence of a law or norm regarding transparency (yes/no)	Society
5	Employment and economic competitiveness	Innovation	Number of collaborations with service providers that comply with applicable sustainability environmental and social criteria	Society

3.2.9 Plastics - Polylactic acid

Polylactic acid (PLA) is a bio-based thermoplastic aliphatic polyester, with lactic acid as its building block. Lactic acid is produced by bacterial fermentation of carbohydrates or chemical synthesis, although the former predominates. Different starchy, saccharic and lignocellulosic crops can serve as a feedstock. PLA is suitable for further processing to create a variety of different products through common plastic processing technologies, such as injection moulding or hot press moulding. PLA-based products are biodegradable under controlled composting conditions. Depending on the desired properties, PLA could substitute fossil-based (non-) biodegradable plastics. The stepwise consistency check carried out for this case study is presented in Table 21. The identified highly relevant indicators are listed in Table 22.

Table 21: Consistency check for the Plastics - Polylactic acid case study

Methodological step	Additional methodological remarks & results
1. Filtering of societal objectives	SO 1. Food and nutrition security was eliminated after selecting “Product” as a scale (Step 5). Food security, however, is a social and ethical topic that is highly debated when it comes to the production of bioplastics from biomass that can be consumed as food. According to European Bioplastics e.V., less than 0.02% of global agricultural areas is currently used as feedstock for the production of bioplastics. For this reason, a deselection of “Product” on the scale level was not undertaken and SO 1. was not considered. Further down the line, however, it may be important to reconsider this eliminated SO. The following four SOs were found relevant for this case study: - SO 2. Sustainable natural resource management - SO 3. Dependence on non-renewable resources - SO 4. Mitigating and adapting to climate change - SO 5. Employment and economic competitiveness
2. Checking completeness societal objectives	No additional SOs were identified.
3. Filtering of sub-topics	A set of sub-topics was considered irrelevant in the context of the bioplastics case study, and hence was excluded from further analysis. These comprise of the following: - From SO 3. Dependence on non-renewable resources: <i>Marine environments' quality.</i> (the marine environments' quality was excluded because the indicators from this sub-topic that are relevant to the PLA case study are already mentioned in the sub-topic water quality of the same SO) - From SO 2. Sustainable natural resource management: <i>Resilience of ecosystems, and Sustainability threshold levels for Bioeconomy Technologies.</i> The sub-topics that were found highly relevant can be found in Table 22.
4. Checking completeness sub-topics	No additional sub-topics were identified.
5. Choosing a scale	The PLA case study links to a specific value chain, covering the processing of various biomass feedstocks to produce the bioplastic. Therefore, it was decided to adopt the indicators linked to the product-level, and to deselect the macro-level (i.e., country or regional level) focused indicators in the database.

6. Ranking of indicators	From the filtering carried out in the earlier steps, 164 indicators were left out of a total initial list of 264. A literature review and scanning of deliverables from sister projects were performed in order to assess the environmental, economic and societal sustainability aspects of the PLA case study. Based on this search, the indicators in the database were scored in terms of their relevance, resulting in: - 35 indicators of high relevance, - 98 indicators of low relevance, - 23 indicators of no relevance, - 8 indicators where the relevance is unknown. From the 35 indicators labelled as highly relevant, most indicators are covered by the environmental dimension (27), followed by the societal (5), and economic (2) dimension. In addition, a single highly relevant indicator was identified as a combination of dimensions: societal and economic (1). An overview of the indicators considered highly relevant can be found in Table 22.
7. Checking completeness indicators	No additional indicators were identified.

Table 22: Highly relevant indicators identified for the Plastics - Polylactic acid case study

SO	Societal objectives	Sub-topics	Indicators	Domains
2	Sustainable natural resource management	Biodiversity	Biodiversity damage potential (BDP; m2*year*PAS (potentially affected species)	Environment
2	Sustainable natural resource management	Land use change (LUC)	Agricultural land converted to energy (or bio-based product) crop	Environment
2	Sustainable natural resource management	Land use change (LUC)	Forest converted to cropland	Environment
2	Sustainable natural resource management	Sustainable resource use	Amount of water used in the value-chain	Environment
2	Sustainable natural resource management	Sustainable resource use	Water consumption (use) (m3/product unit processed)	Environment
2	Sustainable natural resource management	Sustainable resource use	Amount of organic or mineral fertilizers and pesticides used (kg)	Environment

2	Sustainable natural resource management	Sustainable resource use	Intensity of fossil fuel use (m3/ton of product)	Environment
2	Sustainable natural resource management	Sustainable resource use	% of carbon content derived from renewable raw materials	Environment
2	Sustainable natural resource management	Energy efficiency	Renewable energies (RES) share in final energy consumption / product unit	Environment
2	Sustainable natural resource management	Water quality	Ecotoxicity for aquatic fresh water (Comparative Toxic Unit for ecosystems)	Environment
2	Sustainable natural resource management	Water quality	Acidification (mol H+ eq from NOx, SOx, NH3)	Environment
2	Sustainable natural resource management	Water quality	Eutrophication (gr eq PO4)	Environment
2	Sustainable natural resource management	Soil quality	Soil erosion / product unit	Environment
2	Sustainable natural resource management	Forest quality	% of the total amount of wood used comes from sustainable forest management (economically viable, environmentally sound, socially responsible) or are waste wood according to waste wood categories	Environment
2	Sustainable natural resource management	Forest quality	% of deforestation or afforestation /product unit	Environment
2	Sustainable natural resource management	Hazardous substances management	% of hazardous (carcinogenic, mutagenic, reprotoxic) substances	Environment
2	Sustainable natural resource management	Hazardous substances management	% of product materials produced and managed to high environmental and social standards	Environment
2	Sustainable natural resource management	Waste generation and management	Waste generated	Environment
2	Sustainable natural resource management	Waste generation and management	Declaration of end-of-life disposal (in the CSR report) (yes/no)	Environment
2	Sustainable natural resource management	Waste generation	% of materials derived from any other biomass by-products	Environment

		and management		
2	Sustainable natural resource management	Waste generation and management	Recycling rate/cascading indexes	Environment
2	Sustainable natural resource management	Waste generation and management	% of biodegradability	Environment
2	Sustainable natural resource management	Waste generation and management	Post-consuming recycling rate/cascading indexes	Environment
3	Dependence on non-renewable resources	Material use efficiency	Biomass Utilisation Factor (BUF)	Environment
4	Mitigating and adapting to climate change	Greenhouse gas emissions	Life cycle GHG emissions (gr eq. CO ₂ /product unit)	Environment
4	Mitigating and adapting to climate change	Greenhouse gas emissions	Global Warming Potential (GWP100; gr eq CO ₂)	Environment
4	Mitigating and adapting to climate change	Air quality	Amount of volatile organic compounds (VOC) emitted (parts per billion (ppb), parts per million (ppm), or as micrograms per cubic meter (µg/m ³))	Environment
5	Employment and economic competitiveness	Bioeconomy-driving Policies	Strategy for creating the infrastructure and systems necessary for recovering and recycling materials (yes/no)	Economy
5	Employment and economic competitiveness	Value Added of the bioeconomy sectors	Contribution of the product/service/organization to economic progress (revenue, gain, paid wages, R+D costs in relation to revenue, etc.)	Economy
5	Employment and economic competitiveness	Bioeconomy-driving Policies	Presence of a formal policy in place concerning health and safety of all workers, including contractors (yes/no)	Society
5	Employment and economic competitiveness	Stakeholder engagement and capacity building	Presence of a law or norm regarding transparency (yes/no)	Society
5	Employment and economic competitiveness	Employment	Full time equivalent jobs along the full value chain of a product	Society, Economy

5	Employment and economic competitiveness	Employment	Risk of forced labour used for production of commodity (yes/no)	Society
5	Employment and economic competitiveness	Employment	Presence of working children under the legal age (yes/no)	Society
5	Employment and economic competitiveness	Employment	Number of innovative social project that positively impacts the local community	Society

3.2.10 Plastics - Polypropylene from used cooking oil

Polypropylene (PP) represents one of the most widely used plastics in Europe thanks to its versatility. Its bio-based version can be synthesised through three routes, one being the hydrotreatment of used cooking oil (UCO). UCO is converted into high-value hydrotreated vegetable oil (HVO). From this process, a renewable HVO naphtha-grade product is obtained. This case study focuses on the cracking of this “bio-naphtha” to obtain propylene via a process equivalent to petrochemical steam cracking. As such, a bio-based PP is retrieved, which is identical to petrochemical PP. The stepwise consistency check carried out for this case study is presented in Table 23. The identified highly relevant indicators are listed in Table 24.

Table 23: Consistency check for the Plastics - Polypropylene from used cooking oil case study

Methodological step	Additional methodological remarks & results
1. Filtering of societal objectives	SO 1. Food and nutrition security was excluded from the filtering process in Excel, since the feedstock of this case study is considered a waste and, therefore, not related to food applications. Accordingly, the following four SOs were found relevant: - SO 2. Sustainable natural resource management - SO 3. Dependence on non-renewable resources - SO 4. Mitigating and adapting to climate change - SO 5. Employment and economic competitiveness
2. Checking completeness societal objectives	Transforming an existing industry into a circular one requires fulfilling conditions linked to good governance and compliance with the legislation. These aspects could be covered by the readily proposed “Risk assessment and monitoring” SO (SO8, Table 2). Finally, the petro-chemical industry has traditionally been more represented by men than women (gender discrimination, equal pay, etc.) Such aspects can be relevant in the PP sector, and could be covered by the readily proposed “Just inclusive and resilient economy” SO (SO7, Table 2).

3. Filtering of sub-topics	As the present case study does not directly involve any biomass cultivation, processing or trading activities, a set of sub-categories linked to agro-industrial production systems were excluded. These include the following: - From SO 2. Sustainable natural resource management: <i>Biodiversity, Indirect Land use change (iLUC), Land cover/use, Land use change (LUC), Primary biomass production, Protected areas, Resilience of ecosystems, Soil quality, Forest quality and Marine environments' quality.</i> - From SO 3. Dependence on non-renewable resources: <i>Biomass self-sufficiency rate, and Certified bio-based products.</i> The sub-topics that were found highly relevant can be found in Table 24.
4. Checking completeness sub-topics	As a sub-category specifically linked to the economic prosperity or profitability of a production system, a new sub-topic was added: - "Economic viability", to SO 5. Employment and economic competitiveness. Additionally, the sub-topic of "Gender equality" linked to SO7 Just inclusive and resilient economy, is found relevant.
5. Choosing a scale	PP from UCO is linked to a specific value chain. This study focuses on the cracking of renewable HVO naphtha grade product to obtain propylene via a process equivalent to petrochemical steam cracking. Therefore, it was decided to adopt the indicators linked to the product-level. Nevertheless, given the abundance of used cooking oil in the EU and, specially, the importance of PP, the impact of upscaling the process within the case study had to be considered. Accordingly, the macro-level focused (i.e., country and regional level) indicators in the database were selected too.
6. Ranking of indicators	From the filtering carried out in the earlier steps, 113 indicators were selected out of a total initial list of 264. Based on expert judgement, the indicators in the database were scored in terms of their relevance, resulting in: - 31 indicators of high relevance, - 39 indicators of low relevance, - 16 indicators of no relevance, - 27 indicators where the relevance is unknown. From the 31 indicators labelled as highly relevant, most indicators are covered by the environmental dimension (21), followed by the economic (5), and societal (5) dimension. In addition, a single highly relevant indicator was identified as a combination of dimensions: economic, environmental and circular (1). An overview of the indicators considered highly relevant, can be found in Table 21.
7. Checking completeness indicators	From the review of the additional indicators linked to the proposed 3 SOs (Tables 2 and 3), 17 additional product-level indicators were identified to be relevant of which 2 indicators were ranked to be highly relevant, both linked

	to the Society dimension. In addition, 3 indicators were included from the added sub-topic “Economic viability”, and considered highly relevant. These 5 additional indicators are marked in red in Table 21. Accordingly, a total of 36 indicators were marked as highly relevant.
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Table 24: Highly relevant indicators identified for the Plastics - Polypropylene from used cooking oil case study

#	Societal objectives	Sub-topics	Indicators	Domains	Scale
2	Sustainable natural resource management	Sustainable resource use	Abiotic depletion (kg Sb eq.)	Environment	Product
2	Sustainable natural resource management	Sustainable resource use	Resource depletion (mineral, fossil) / product unit (MJ)	Environment	Product
2	Sustainable natural resource management	Sustainable resource use	Change in consumption level of fossil fuel resources / product unit	Environment	Product
2	Sustainable natural resource management	Waste generation and management	Waste generated	Environment	Product
2	Sustainable natural resource management	Waste generation and management	Procedures for separating and using recyclable materials from the waste stream (yes/no)	Environment	Product
2	Sustainable natural resource management	Waste generation and management	Procedures for recovering materials for other uses, such as incineration for raising process steam or heating, or agricultural use (yes/no)	Environment	Product
2	Sustainable natural resource management	Waste generation and management	% of product is actively being recovered and recycled	Environment	Product
2	Sustainable natural resource management	Waste generation and management	% of by-products or wastes reused by the processing/production facility or transferred to other sectors	Environment	Product
2	Sustainable natural resource management	Waste generation and management	% of biodegradability	Environment	Product
2	Sustainable natural resource management	Waste generation and management	Procedures for handling, collecting, separating and disposal of hazardous waste as defined by the relevant local and national regulatory authorities (yes/no)	Environment	Product
3	Dependence on non-renewable resources	Bio-energy replacing non-renewable energy	% of biofuels produced with co-products, residues and waste	Environment	Product

3	Dependence on non-renewable resources	Bio-energy replacing non-renewable energy	Biofuels	Environment	Country
3	Dependence on non-renewable resources	Bio-energy replacing non-renewable energy	Municipal waste	Environment	Country
3	Dependence on non-renewable resources	Bio-material replacing non-renewable resources	Bio-based plastics	Environment	Country
3	Dependence on non-renewable resources	Material use efficiency	Material and (municipal) waste recycling and recovery rates	Economy, environment, circularity	Country
4	Mitigating and adapting to climate change	Greenhouse gas emissions	Life cycle GHG emissions (gr eq. CO ₂ /product unit)	Environment	Product
4	Mitigating and adapting to climate change	Greenhouse gas emissions	Rate of GHG emissions reduction or savings	Environment	Product
4	Mitigating and adapting to climate change	Greenhouse gas emissions	Global Warming Potential (GWP100; gr eq CO ₂)	Environment	Product
4	Mitigating and adapting to climate change	Air quality	Ozone Depletion Potential (ODP)	Environment	Product
4	Mitigating and adapting to climate change	Climate footprint	Climate footprint	Environment	Country
4	Mitigating and adapting to climate change	Greenhouse gas emissions	Greenhouse gas emissions	Environment	Country
5	Employment and economic competitiveness	Innovation	Presence of environmentally/friendly technologies (yes/no)	Environment	Product
3	Dependence on non-renewable resources	Material use efficiency	Recycling rate of bio-based products	Economy	Country
5	Employment and economic competitiveness	Bioeconomy-driving Policies	Strategy for creating the infrastructure and systems necessary for recovering and recycling materials (yes/no)	Economy	Product
5	Employment and economic competitiveness	Comparative advantage	Potential market share of the product	Economy	Product

5	Employment and economic competitiveness	Comparative advantage	Number of partnerships in R&D	Economy	Product
5	Employment and economic competitiveness	Economic viability	Return on Investment (ROI)	Economy	Product
5	Employment and economic competitiveness	Economic viability	Cost of production (COP)	Economy	Product
5	Employment and economic competitiveness	Economic viability	Cost-competitiveness with the conventional, incumbent product	Economy	Product
5	Employment and economic competitiveness	Stakeholder engagement and capacity building	Number of informal workshops to build local understanding in the community of the processes that may impact them directly to aid meaningful engagement	Society	Product
5	Employment and economic competitiveness	Employment	Adoption of Corporate Social Responsibility (CSR) certificate (yes/no)	Society	Product
5	Employment and economic competitiveness	Innovation	Involvement in technology transfer program or projects (yes/no)	Society	Product
5	Employment and economic competitiveness	Bioeconomy-driving Policies	Country level strategies	Society	Country
5	Employment and economic competitiveness	Employment	Quality of employment	Society	Country
7	Just inclusive and resilient economy	Gender equality	Number of women having equal job opportunities as men	Society	Product
7	Just inclusive and resilient economy	Gender equality	Ratio of basic salary of men to women by employee category	Society	Product

3.3 General feedback on the bioeconomy monitoring and assessment framework

Besides the case study specific evaluation, case study owners also had some general feedback for the draft bioeconomy monitoring and assessment framework for further improvement. This information is collected and categorised into different topics and is presented in Table 25. The term ‘element’ in the table may refer to one of the levels in the framework (i.e., dimension, SO, sub-topic or indicator).

Table 25: Consolidated and categorised general feedback on the bioeconomy monitoring and assessment framework

Type of comment	Applicability - level of framework	Applicability - further specification	Description of the comment
Missing/ recategorizing of element	Sub-topic	Legal and social compliance	Within the social domain, a sub-topic dedicated to aspects such as codes of conduct, due diligence and compliance with legislation is missing. Therefore, a new sub-topic "Legal and social compliance" can be added, which could f.i., cover indicators 217-221, 228, 232, 233, and 239.
Missing/ recategorizing of element	Sub-topic	Sustainable forestry	Within the existing sub-topic "Forest quality", certain aspects related to sustainable forest management and the maintenance of forest productivity are missing, such as harvest not exceeding regeneration rates. Therefore, it is proposed to add a more general sub-topic "Sustainable forestry", which covers all forest-specific aspects (e.g., including indicators 87-92 from the original "Forest quality" sub-topic).
Missing/ recategorizing of element	Sub-topic	Water quality and quantity	Within SO 2. Sustainable natural resource management, indicators linked to water quality and quantity are scattered across sub-topics. Therefore, it is proposed to add a new sub-topic "Water quality and quantity", which covers all water-related indicators.
Missing element	Indicator	Use of resources sourced from abroad	Within the framework, an indicator that monitors the usage of resources sourced from abroad, as well as the related impacts, is missing. Since the Bioeconomy Strategy explicitly mentions "reducing dependence on non-renewable, unsustainable resources whether sourced domestically or from abroad" as its third policy objective, it is proposed to add an indicator on this aspect.
Missing element	Indicator	Additional indicators	A compilation of indicators linked to the environmental, social and economic sustainability of biobased products at the product level was made by WR in a previous EU project (EC (2022), Appendix E-3). As some of the indicators are not yet covered in the present draft monitoring and assessment framework, these could be used to complement it.
Element is too narrow	Sub-topic	Hazardous substances management	Within this sub-topic, an indicator is missing about the leaching of microplastics to the environment. Hence, it is proposed to extend this sub-topic with an additional indicator on this aspect.

Element is too narrow	Sub-topic	Sustainable resource use/ Energy efficiency	Within one of these sub-topics, an indicator is missing on the renewable energy self-sufficiency of a process(ing facility). Hence, it is proposed to extend one of these sub-topics with an additional indicator on this aspect.
Element is too narrow	Sub-topic	5. Employment and economic competitiveness	Within this SO, an indicator is missing on (mitigated) CO ₂ emissions per invested or gained euro. Such an indicator offers a measure of the degree of future-proofness of an organization or value chain, and progress towards achieving sustainability goals. Hence, it is proposed to extend this SO with an additional indicator on this aspect.
Element is too narrow	Indicator	Indicator 79	AOX is a very specific substance and there are numerous other important wastewater pollutants (such as heavy metals, which are currently not represented in the monitoring framework). Which pollutant is relevant in a particular case strongly depends on the specific production process. Generally, wastewater pollutants are classified as either “organic” or “inorganic”, with total organic carbon (i.e., TOC) forming a widespread method for organic pollution detection. Therefore, for the sake of simplicity, it is proposed to replace the indicators on AOX by two new, aggregated indicators, being: - “Inorganic pollutants in water” and - “Total organic carbon (TOC) in water”. When opting for one of these indicators, users of the monitoring framework will be asked to additionally specify which substance(s) is, or are, relevant to the examined product system. This issue also applies to the sub-topics “Air quality” and “Soil quality”, and requires further discussion.
Unclear element	Sub-topic	Sustainability threshold levels for Bioeconomy Technologies	From the name of this sub-topic, it cannot be derived directly what it entails, and whether the indicators it includes are already covered by other sub-topics within SO 2. Sustainable natural resource management. It is proposed to clarify this sub-topic, and reconsider its scope if needed.

Unclear element	Indicator	Indicators 23, 28, 45, 50, 62, 69, 100, 101, 110, 115, 118, 123, 129, 143, 146, 148-149, 151, 155 184, 199, 201, 207, 209, 215, 217, 218, 224, 227, 228, 230, 236, 238-240, 248-250, 252, 262, 265	Each of these indicators are considered to be unclear for one of the following reasons: - One or more terms in the indicator are not clearly defined, or - It is unclear to what aspects the indicator applies, or - It is not clear how the indicator is measured. <i>For example, indicator 69 demands the "% of biological content/product". However, it is not clear how the term "biological" is defined and how it should be measured.</i>
Double/ overlapping elements	Domain	Environmental	Some of the indicators are repeated under different sub-topics, while it is understandable how that came to be, it would be better to categorise indicators that are overarching and fall under more than one sub-topic or even SO.
Double/ overlapping elements	SO	2. Sustainable natural resource management 3. Dependence on non-renewable resources 4. Mitigating and adapting to climate change	There are overlaps between several indicators and sub-topics between SO 3 with SO 2 and SO 4. For example, is it necessary to collect data on bioenergy if in SO 4 GHG-emissions are evaluated? The share of non-renewable resources replaced by biomass alone is not informative in regard to sustainability, as the use of biomass can have both positive and negative effects (which are already covered by other sub-targets or indicators such as GHG emissions, iLUC, biodiversity etc.). I guess the key question here is whether the use of biomass helps reduce the use of non-renewable resources. Maybe therefore the relevant sub-topic/indicators are simply the non-renewable resource use?
Double/ overlapping elements	Sub-topic	Bio-material replacing non-renewable resources	This subtopic appears to be redundant since a separate indicator is dedicated to the bio-based content of a product (i.e., indicator 174 "Bio-based content"). Therefore, it is proposed to consider removing this sub-topic, if not at the expense of valuable information.
Double/ overlapping elements	Indicator	Indicators 59 & 63; 61 & 65 & 66 & 67; 71 & 72; 100 & 153; 132 & 133; 143 & 144 & 145; 192 & 193 & 195 & 196	The indicators within each of these indicator pairs seem to overlap, or it is not clear in which respect the individual indicators differ. <i>For example, indicators 59 "Amount of water used in the value-chain" and 63 "Water consumption (use) (m3/product unit processed)" both provide a measure of the water consumed in the value chain. Therefore, it is proposed to consider removing one indicator for each indicator pair, if not at the expense of valuable information.</i>

Double/ overlapping elements	Indicator	Indicator 183	This indicator, which measures "Carbon intensity" seems to overlap with SO 4 Mitigating and adapting to climate change. Therefore, it is proposed to consider removing this indicator, if not at the expense of valuable information.
Double/. overlapping elements	Indicator	Indicator 39	This indicator, which measures "Wood consumption", seems to overlap with several indicators covered by the sub-topic "Waste generation and management" (e.g., indicator 132 and 136), as it is unclear if it also relates to wood residues or biomass by-products. Therefore, it is proposed to clarify this indicator and remove any overlaps, if not at the expense of valuable information.
Proposed recategorisation of an element	SO	2. Sustainable natural resource management	The SO 2. Sustainable natural resource management covers a variety of aspects, which may confuse users of the monitoring and assessment framework. To reduce complexity and avoid confusion, it is proposed to limit the coverage of SO 2. to sub-topics that represent actual natural resources (i.e., soil, water, air). In addition, it could be considered to also include biomass- (and waste-) related sub-topics, as well as biodiversity as a cross-medium natural resource. All other sub-topics (e.g., linked to iLUC, energy and GHG emissions) could be moved to the indicator column, as these are the effects of changes in the sphere of natural resources.
Proposed recategorisation of an element	Sub-topic	Air quality	The sub-topic "Air quality" is at present covered by SO 4. Mitigating and adapting to climate change. However, it is usually categorized separately from climate change, as it refers to the pollutants in the air, like particulate matter, SOx and NOx, which are all greenhouse gases. Therefore, it is proposed to move this sub-topic to SO 2. Sustainable natural resource management, alongside the water and soil quality sub-topics.
Proposed recategorisation of an element	Sub-topic	Indirect Land use change (iLUC)	The sub-topic iLUC is at present covered by SO 2. Sustainable natural resource management. It is proposed to consider moving this sub-topic to SO 1. Food and nutrition security.

Proposed recategorisation of an element	Indicator	Indicators 73, 87-89, 99, 110, 116, 123, 152, 154, 194	These indicators possibly fit better in <i>another sub-topic</i> . <i>For example, indicator 99 "% of plastic (synthetic plastic material, plasticizers, PVC) contained in the product" is currently assigned to the sub-topic "Hazardous waste management". However, plastic is not per definition a hazardous material and can in certain instances be recovered, so instead this indicator may fit better to the sub-topic of "Sustainable resource use" or "Waste generation and management". Therefore, it is proposed to consider moving the indicator to the better-fitting sub-topic.</i>
Proposed recategorisation of an element	Indicator	Indicators 123-140, 208	These indicators possibly fit better in another (or an additional) domain. <i>For example, indicator 127 "% of the product is actively being recovered and recycled" is currently labelled as an environmental indicator, whereas it is also relevant in the domain of circularity.</i> Therefore, it is proposed to consider labelling these indicators by the additionally relevant domains.
Proposed renaming of an element	Sub-topic	Energy efficiency	The sub-topic "Energy efficiency" at present covers several indicators that are not related to efficiency measures. Therefore, it is proposed to modify the name of this sub-topic to "Energy consumption", and consider "Energy efficiency" as an indicator within this sub-topic.
The element is too broad	Indicator	Indicators 213, 214, 231, 239	These indicators comprise of, or measure, different aspects. <i>For example, indicator 213 "Number and variety of relevant stakeholders participating in the consultative process" demands two data inputs, both including the number and variety of relevant stakeholders, while being formulated as a single indicator.</i> Therefore, it is proposed to consider splitting them into two or more different indicators.

3.4 Insights generated from the validation of the bioeconomy monitoring and assessment framework

Here we present key observations from the consistency check results:

- In total, five case study owners indicated the need to include additional SOs linked to these societal impacts. These are the Recycled carded wool from Prato Italy, Man-made cellulosic fibres, Latxa sheep wool from the Basque country of Spain, Chemicals from waste, and Polypropylene from used cooking oil case studies. These case studies indicate gaps in terms of societal objectives dealing with human health, gender balance, risk assessment and good governance. This shows that there are some topics that are not yet sufficiently represented in the draft framework and the framework can be expanded to also include these aspects. Overall, the additional SOs that were proposed to be included in the draft of the bioeconomy monitoring and assessment framework are:
 - SO6: Disease prevention and human health.
 - SO7: Just inclusive and resilient economy.
 - SO8: Risk assessment and monitoring.
 - SO9: Promoting global governance and collaboration.

- Concerning the existing 5SOs (SO1-SO5) and the newly proposed 4SOs (SO6-SO9), several case study owners proposed to enrich the monitoring and assessment framework with new sub-topics linked to one or multiple sustainability dimensions. This resulted in the inclusion of an additional 19 sub-topics (in addition to the 39 sub-topics in the original framework). These include:
 - Social dimension (13):
 - Linked to SO6 “Human health”, “Disease/hazards prevention” and “Food safety”;
 - Linked to SO7 “Resilience of rural communities - social protection”, “Gender equality”, “Labour rights”, “Inclusiveness”, “Respect of indigenous rights and land rights”, “Access to basic needs” and “Physical infrastructure”;
 - Linked to SO8 “Monitoring and accountability systems” and “Risk assessment”;
 - Linked to SO9 “Partnerships and cooperation”.
 (by Latxa, man-made cellulose, Prato, chemicals from waste and PP cases)
 - Economic dimension (3): Linked to SO5 “Externalities”, “Environmental costs” and “Economic viability” (by Latxa, Prato, chemicals from waste and PP cases)
 - Environmental dimension (3): “Use of local resources”, “Local production” (by cross-laminated timber case) and “Supply security” (by Prato case).

It is seen that these additional sub-topics are predominantly linked to the social dimension showing that this dimension is not yet well represented in the draft of the bioeconomy monitoring and assessment framework.

- Also with the inclusion of the additional SOs and sub-topics as summarized above, 109 additional indicators were proposed to be included in addition to the 264 indicators inventoried in the draft of the bioeconomy monitoring and assessment framework.
 - 38 indicators linked to the SOs 6, 7 and 8 at the product-level (FAO: Bracco et al., 2019);
 - 34 S-LCA and LCC indicators to extend the social and economic dimensions at the product level (ORIENTING project, 2022);
 - 28 regional level indicators to extend the territorial level assessment (ISTAT, 2023);

- 9 individual indicators proposed additionally from case studies that are not yet covered and found relevant from literature review (laquaniello et al., 2017, Iribarren et al., 2022).
- In the majority of case studies, SO 1. Food and nutrition security was excluded from the analysis at the first step (Step 1. Filtering of the SOs). This is due to the fact that these case studies did not use biomass as feedstock that competes with food. For the Polylactic acid case study, SO1 was found relevant but got filtered out later on (Step 5. Choosing a scale). This is because the indicators proposed for this SO1 were all linked to the macro level and the product level indicators were not present. Accordingly, SO1 was not sufficiently analysed and checked as part of this exercise with the case studies. It is therefore recommended to consider adding some product-level indicators for SO1 since this topic is of high ethical/social relevance.
- The number of highly relevant indicators reported by the different case studies varies significantly, from 23 (in the TERNUA case) up to 97 (in the MMCF case). This might (in part) be attributed to the differences in the background knowledge of each case study owner on the case study at hand. It seems at first glance that case study owners who had a deeper knowledge of the case study content (e.g., from previous studies, or by consulting external experts, or by performing a literature review), marked a substantially lower number of indicators as highly relevant than the case studies which were analysed with the expert judgement of the case study owners based on preliminary information on the case study.
- Moreover, it appears there are substantial differences among case studies in terms of how the indicators were ranked based on their relevance (i.e., high, medium, low or no). One possible explanation for this outcome is the strong sector dependence of indicators (e.g., the indicator “Biobased textiles” specifically applies to the textile sector). However, several indicators were assigned a high level of relevance in either all or the majority of case studies. Considering these frequently recurring, highly relevant indicators, five different categories consisting of similar indicators could be distinguished. These include:
 - Greenhouse gas emissions and global warming.
 - Waste generated.
 - Use of fossil resources.
 - Recycling and/or reuse of materials.
 - Potential market share of the product.

These indicator categories could be regarded as a common denominator across all case studies. Yet, this does not imply that all other indicators are less relevant. After all, the level of relevance of each indicator is to be examined in the context of the case study and the sector it pertains to.

4. GUIDANCE FOR CASE STUDY ANALYSIS

This chapter provides guidance to the stakeholder mapping, relevant policy mapping and information and data gathering activities to be performed in the case study analysis in T3.3. It

also provides information on the linkages with the WP2, 4 and 5 which the case study analysis will provide input for.

4.1 Mapping stakeholders

For each case study a stakeholder mapping will be performed. This should serve different purposes, among which, (1) it should provide an overview of all stakeholders involved directly and indirectly in the biomass value chain from biomass sourcing, to end of life; (2) it should help identify relevant stakeholders for implementation of interviews to collect information needed; (3) it will help to identify the relevant stakeholders to be invited to the SUSTRACK Inspire, Design and Implement workshops.

In Figure 2, the CBBE system is presented that can support the identification of a wide number of stakeholders. Central in the system is a biomass delivery chain that starts with biomass production or harvesting, via logistics, pretreatment to conversions to distribution and then to end products and uses. These chains are all based on renewable biological resources and include conversions into biotextiles, bioplastics, biochemicals and building materials. Activities in the bioeconomy system encompass not only activities within the biomass value chain but also the wider industrial environment and the ‘enabling environment’. The wider industrial environment covers aspects such as food and product labelling and promotion, and minimal quality requirements for products which can partly be arranged through different policy measures but also through voluntary certification and agreements between economic actors. As to the enabling environment, it creates the conditions in which the system functions and covers factors such as transport, infrastructure, R&D and regulations (e.g. food safety and environmental regulations). A number of factors that influence activities at the consumer level are also a central part of the system. They cover the product environment and the characteristics of (individual) consumers, both of which determine the consumer’s relationship to bioproducts and how they manage the waste that comes from their consumption.

For the mapping of stakeholders, the system in Figure 2 can be followed in order to identify the different stakeholders involved directly and indirectly in the biomass delivery chain and the wider CBBE environment. The biomass delivery chains are displayed in Figure 2 as linear while they are in fact more circular as end of life of bio-based products are reused as biomass resources in a (new) biomass delivery chain.

Less relevant for the stakeholder mapping, but very relevant in relation to the development of the bioeconomy monitoring and assessment framework, are the outcomes different activities in the bioeconomy system have within the system in socio-economic, environmental and climate terms. These outcomes are feedback loops that occur between parts of the biomass delivery chain (production, processing, distribution and consumption), and from the socioeconomic and environmental outcomes of bio-based product production and consumption (such as on food security and biodiversity impacts) back to that production and consumption. Furthermore, the different feedback loops interact with one another. For example, certain socio-economic outcomes such as consumers’ income lead to increased consumption and more quantity of products to be managed for end-of-life treatment. This interaction could have both positive and negative environmental feedback loops because it can lead to both higher and lower demand for primary resources which may lower pressures on land, water, biodiversity and non-renewable resources.

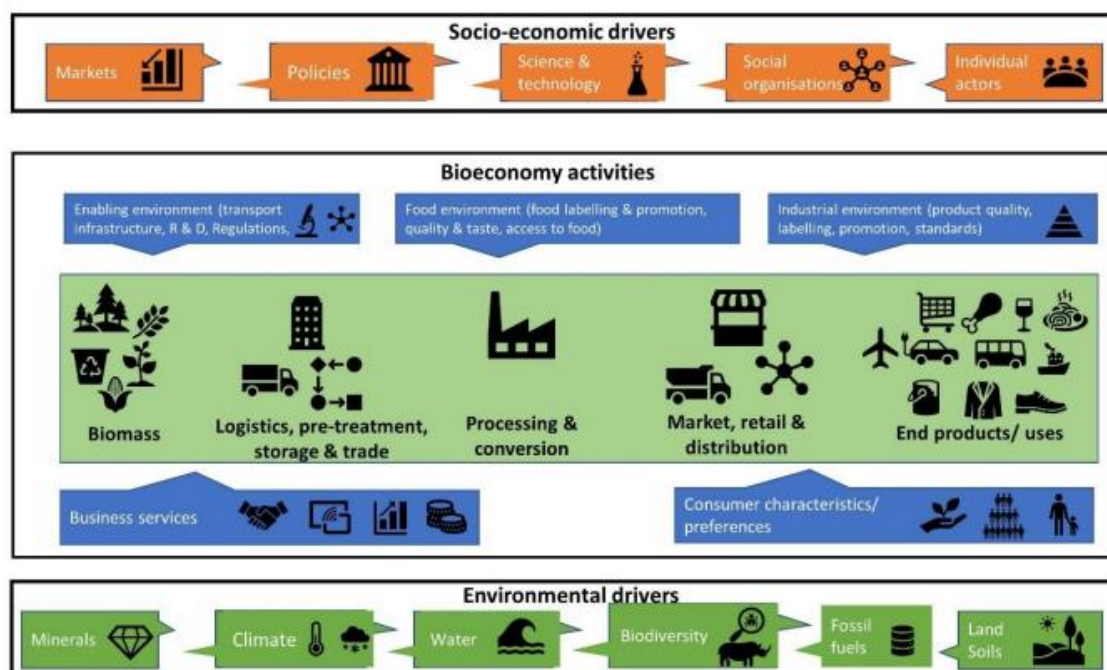


Figure 2. Bioeconomy system made up of different stakeholders

Therefore, for the mapping of stakeholders in the case studies it is important to map stakeholders, companies, that are in the biomass delivery chain, including the marketing and retail and distribution phases and the end-of-life handling. In addition, stakeholders involved in the enabling environment, particularly R&D and policy regulation are also relevant to mapping.

4.2 Mapping policies of relevance

Figure 2 also provides an overview of the different components of the bioeconomy system that can be regulated by policy instruments. These instruments may impact directly the different biomass delivery chain components and the enabling environment, the industrial environment, the business services and the consumer preferences and characteristics within the bioeconomy system. The policy measures may also have a more indirect impact on the bio-based economy system, by regulating the relation between the bioeconomy activities and the socio-economic and environmental drivers. When mapping policy instruments in the case studies, the following types of policy instruments can be identified and described (see also Section 4.4.3 on WP5 needs):

- 1) **Direct regulation** - a command and control approach using obligatory standards and licenses that require people/companies/market players to change their behaviour and punishes them if they are detected to be non-compliant;
- 2) **Economic instruments** - includes all instruments changing price incentives (taxes, subsidies, feed-in tariffs), but also quantity constraints ((tradable) quota, tariff rate quota), and charges. These instruments give people incentives to voluntarily change their behaviour (e.g. based on their own rational cost-benefit calculations);
- 3) **Voluntary approaches** - could be codes of good practice, self-regulation and other industry-led initiatives and certification. Financial incentive schemes could be part of these instruments also coupled with certification. These approaches typically encourage rather than force people or businesses to show the desired behaviour;
- 4) **Information and advice sharing systems** - policies aimed at raising awareness and facilitating changes in behaviour;
- 5) **Market-based signalling approaches** - labelling, traceability, voluntary certification schemes and farm assurance schemes. These approaches are often related to informational problems (lack of information about product quality and food safety) hindering the proper functioning of markets;
- 6) **Other strategies** not in the categories above such as vision documents, road maps, and strategies.

4.3 Information and data collection from literature and stakeholders

To gather information on the case studies, it is essential to conduct a thorough research by scanning relevant articles, previous studies, reports, and other sources of information. This process will involve identifying, screening, and extracting the necessary information. To ensure that relevant literature is not missed, we recommend using a systematic search with main search words in literature databases and collections (e.g. Scopus, Web of Science) and also other internet searches (e.g. Google), but may also include regional reports and publications specifically for the regional case studies. Very helpful sources of information in the context of sustainable transitions are the sustainability strategies that most companies publish on a yearly, 2- or 5-yearly basis.

In addition, information may also be collected through interviews or panel discussions with relevant stakeholders, to collect their views on the main information needs for WP2, 4 and 5 issues and discussed in the following sections.

4.4 Information needs for other Work packages for case studies

4.4.1 Linkage with WP2 related to gap analysis and closing the gaps

Figure 3 provides an overview of the linkage between WP2 and WP3. The preliminary database of indicators for bioeconomy monitoring and assessment from T2.1 is used as input in T3.2 for

validation with case studies and for filling certain research gaps while T3.2 in return provides feedback to WP2.

The insights generated from the consistency checks carried out for each case study are reported which provides information on assessment/indicator gaps to be addressed in T2.2 and for further improvement of the framework in T2.3. The case study assessment of T3.3 will work closely with T2.2 for the development of further indicators/methods that will be tested in case studies.

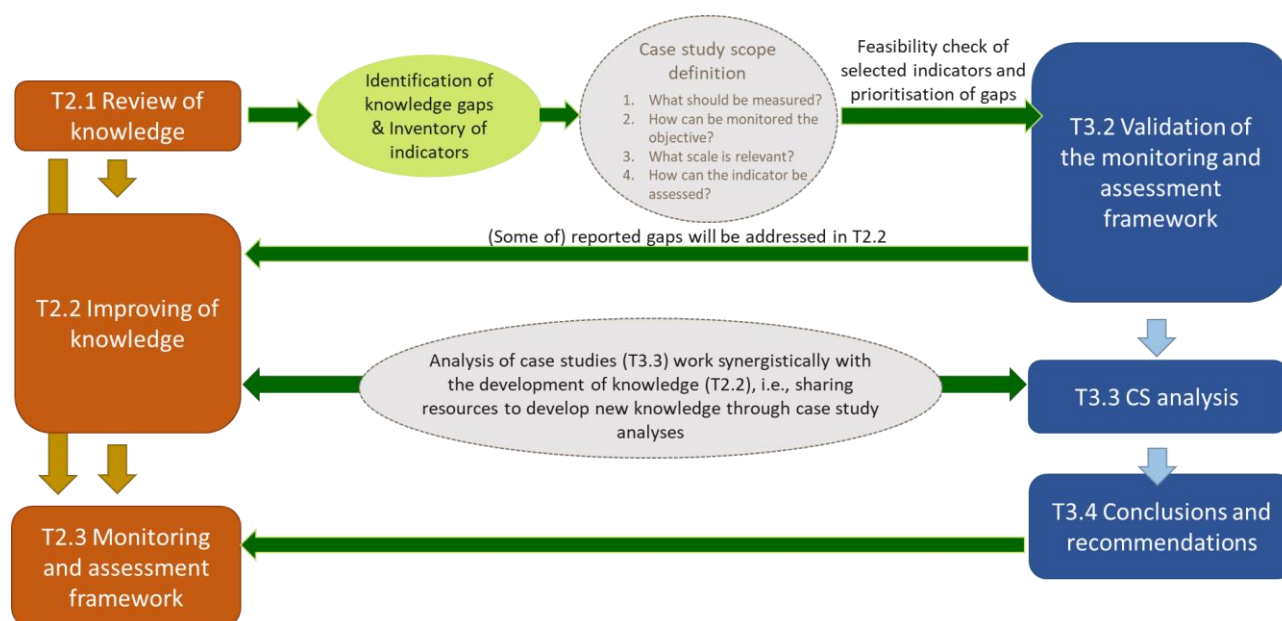


Figure 3: Overview of the linkage of WP2 and WP3 in the Sustrack project

In T2.1 research gaps were identified and key recommendations to further advance bioeconomy monitoring and assessment were derived. In Table 26 the selected research gaps suitable to be addressed at the case study level are presented. During T3.3, a more detailed explanation will be provided regarding the specific research gaps that will be targeted, along with the case studies in which they will be addressed and the methods that will be employed to carry out the research.

Table 26: Research gaps identified in T2.1 to be considered to be addressed in case studies

#	Description
8	Need for better addressing (pure) social aspects related to bioeconomy within the EU bioeconomy framework and the five SOs. Social issues that should be considered may include human health, rural communities' development, equality, inclusiveness etc.
9	Need for a better tailoring of EU monitoring framework for local policy makers. This may include the proposal of sub-national monitoring frameworks taking stock of available data at these lower territorial levels.

10	Need for further standardisation of bioeconomy assessment at the regional-city level (including proposal for monitoring framework, indicators, and methods) to enable comparison among cities/regions.
12	Coverage of governance indicators, which gives insight into how/if stakeholders work together in achieving common strategic goals of a city/local or regional area. E.g. are there cluster organisations, is there cooperation across industries/regions/ministries; are there national/regional/local circular bioeconomy strategic plans.
13	Monetisation of externality: a need for indicators that distinguish between market price and inherent value, e.g., by monetization of environmental externalities. This would permit a fairer comparison between fossil-based and bioeconomy-based systems, supporting decision-makers and investors
15	Impacts on rural communities (at the very local scale) should be considered when evaluating bioeconomy projects as these are the stakeholder category most affected by changes in the bioeconomy value-chains. In this line, the inclusion of governance/collaboration aspects between industries/stakeholders is key.
19	LCA studies should integrate and improve key impact categories for the bioeconomy. These are direct and indirect land-use issues, biodiversity, deforestation, soil and water quality, and related carbon emissions.
20	LCA-based indicators should cover the whole life cycle and not be limited to factory gate boundaries. Above all within construction value-chain studies.

4.4.2 Linkage with WP4 on pathways for changing and defining potential (policy) intervention options

The case study assessments serve as a basis for the analysis of each sector by means of the Green Economy Model (GEM). This includes the development of policy portfolio scenarios for the selected case study countries and sectors. In WP4, simulations will be created for selected countries and case study sectors using GEM, allowing for a better understanding of how the practices analyzed affect overall macroeconomic outcomes, considering their unique characteristics. Insights from the case studies will be used to parameterize interventions in GEM and subsequently to interpret and substantiate the results. These insights will be collected from the case studies in targeted sessions organised as part of the Design and Implement conferences.

4.4.3 Linkage with WP5 on relevant policy instruments

The case studies will support the identification of policy priorities, the definition of policy actions based on discussions on challenges and opportunities, and the subsequent elaboration of general recommendations for the bio-based sector.

As a first step, it is needed to collect all policy instruments in the specific case studies that: (1) support the transition towards more bioeconomy and circularity and (2) are barriers to the sustainable transition. In order to describe these instruments Table 27 specifies the information to be collected:

Table 27: Information to be collected regarding relevant policy instruments

Policy instrument - name	Short description of main goal of the policy instrument	Barrier or stimulation instrument?	How is it a barrier or stimulation instrument?	Geographic coverage (EU, National - which country?)	Timeframe (old, current, future/planned/proposed)	Type of instrument (see Tab type of instruments)	Information sources: link to policy document, or link to policy evaluation documents (if available)

Final policy recommendations and roadmap guidelines will be developed for each case study sector, which will be validated in the case study discussions. These recommendations and guidelines will summarise the findings for each of the carbon-intensive sectors and provide guidance on urgent actions and sustainability considerations to support a sustainable transition at national and EU levels.

4 CONCLUSIONS AND FURTHER STEPS IN WP3

This deliverable supports the testing, validation and further development of the framework for monitoring and assessing the sustainability of the transition to a circular and bio-based economy. It provides results from the consistency check exercise carried out for each of the 10 case studies on the draft framework developed in T2.1. This provided an understanding of the societal objectives, sub-topics and indicators that are missing and/or need further elaboration. This can be used as input on how the monitoring framework can be further improved as part of T2.3. It was seen that especially on the social dimension, some key topics were not yet adequately addressed. This considers human health, gender balance and inclusiveness, labour rights and resilience of rural communities. These were proposed to be addressed with new societal objectives, i.e., SO6 “disease prevention and human health” and SO7 “Just inclusive and resilient economy” and linked new sub-topics and indicators. Additionally, on the economic dimension existing SO5 “Employment and economic competitiveness” was proposed to be further complemented with LCC indicators. Also, aspects linked to good governance and risk assessment measures were found to be relevant to be included. They are proposed to be addressed with new societal objectives i.e. SO8 “Risk assessment and monitoring” and SO9 “Promoting global governance and collaboration” and linked new sub-topics and indicators. Furthermore, the draft framework was found to be lacking indicators for regional level assessment which has been proposed to be included in the further development of the bioeconomy monitoring and assessment framework. From the analysis of identified highly relevant indicators, several indicators were frequently ranked as highly relevant across case studies. These include greenhouse gas emissions, waste generated, use of fossil resources, recycling and/or reuse of materials.

This deliverable, furthermore, provides guidance for the next WP3 task, T3.3 case study assessment in terms of mapping of stakeholders and policies of relevance for case studies, and collection of information and data from literature and stakeholders. Additionally, the linkages

with other SUSTRACK WP activities are defined in a way to underline how the case studies analysis will provide input for the transition to a more circular biobased economy. As a follow-up work, the T3.3 case study assessment will entail testing the environmental, social, and economic sustainability indicators and methodologies selected and developed under the project in terms of applicability, completeness, data availability, and interpretation. Based on the results of this testing, guidance is provided on the performance of the indicators and tools in the specific case study contexts and how they might be improved. The case studies will also be used to test the new methods proposed in T2.2.

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